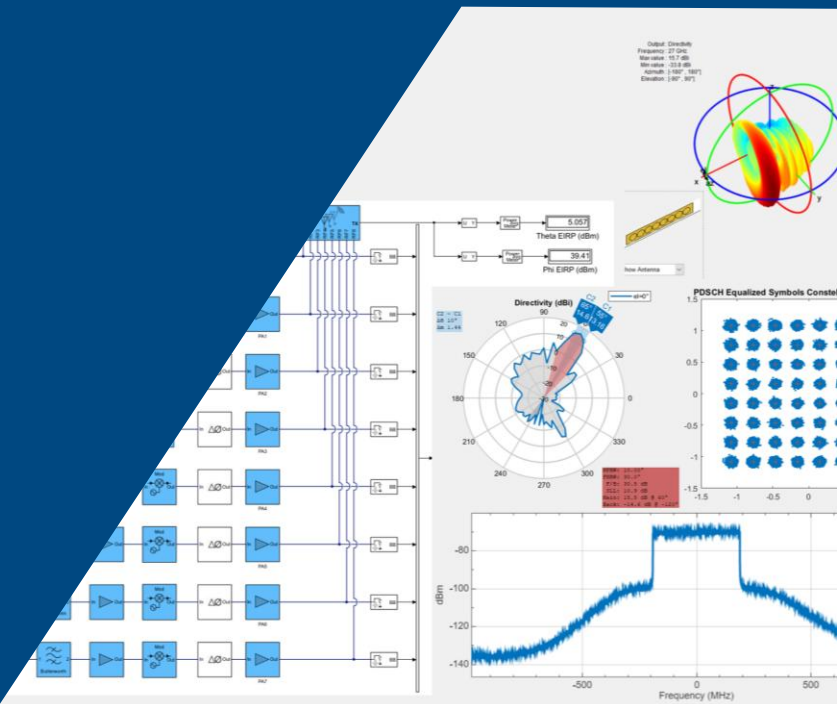
MILCOM 2027 — Tutorial TUT13
AI-Enabled Antenna-to-System Design with LLMs, MCP and MATLAB

Part 3 — Antenna-to-System Integration

Sekhar Sekharan– sekhars@mathworks.com

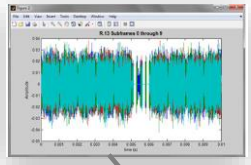
Applications Engineer
MathWorks | RF/EM Group



© 2026 The MathWorks, Inc.

RF Systems Are Complex …

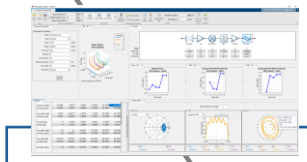
Waveforms



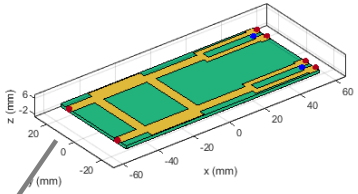
Data conv. Clock gen.



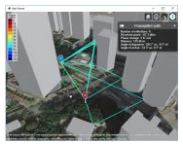
RF Transceivers
Amplifiers



Matching / Feed Networks / Filters



Propagation



Digital Baseband

Mixed-Signal

RF

Antenna

Environment

TX

Mod

DAC

PA

TX

Targets

Channel

DSP

Equal.

ADC

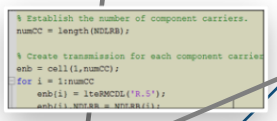
LNA

RX

RX

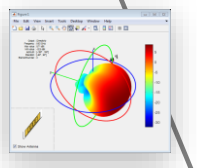
Interferers

Control Logic



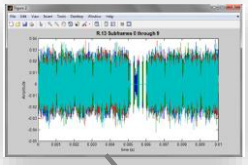
Algorithms / Implementation

T&M / OTA



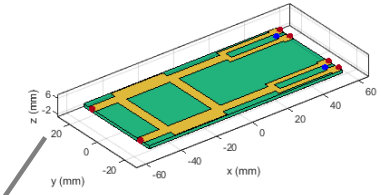
Antenna Arrays

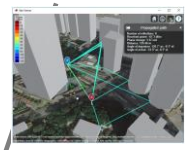
… and Market Trends Make Design Even Harder











Digital Baseband

Mixed-Signal

RF

Antenna

Environment

TX

Mod

DAC

PA

RX

DSP

Equal.

ADC

LNA

#1. Complex Modulations

#2. Large Bandwidth

#3. High Frequency (mmWave, FR3)

#4. More Antennas

Targets

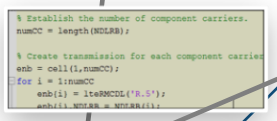
Channel

Interferers

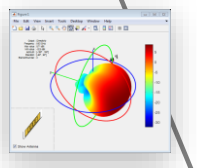
Control Logic

#5. DSP Algorithms Everywhere (AI)

T&M / OTA

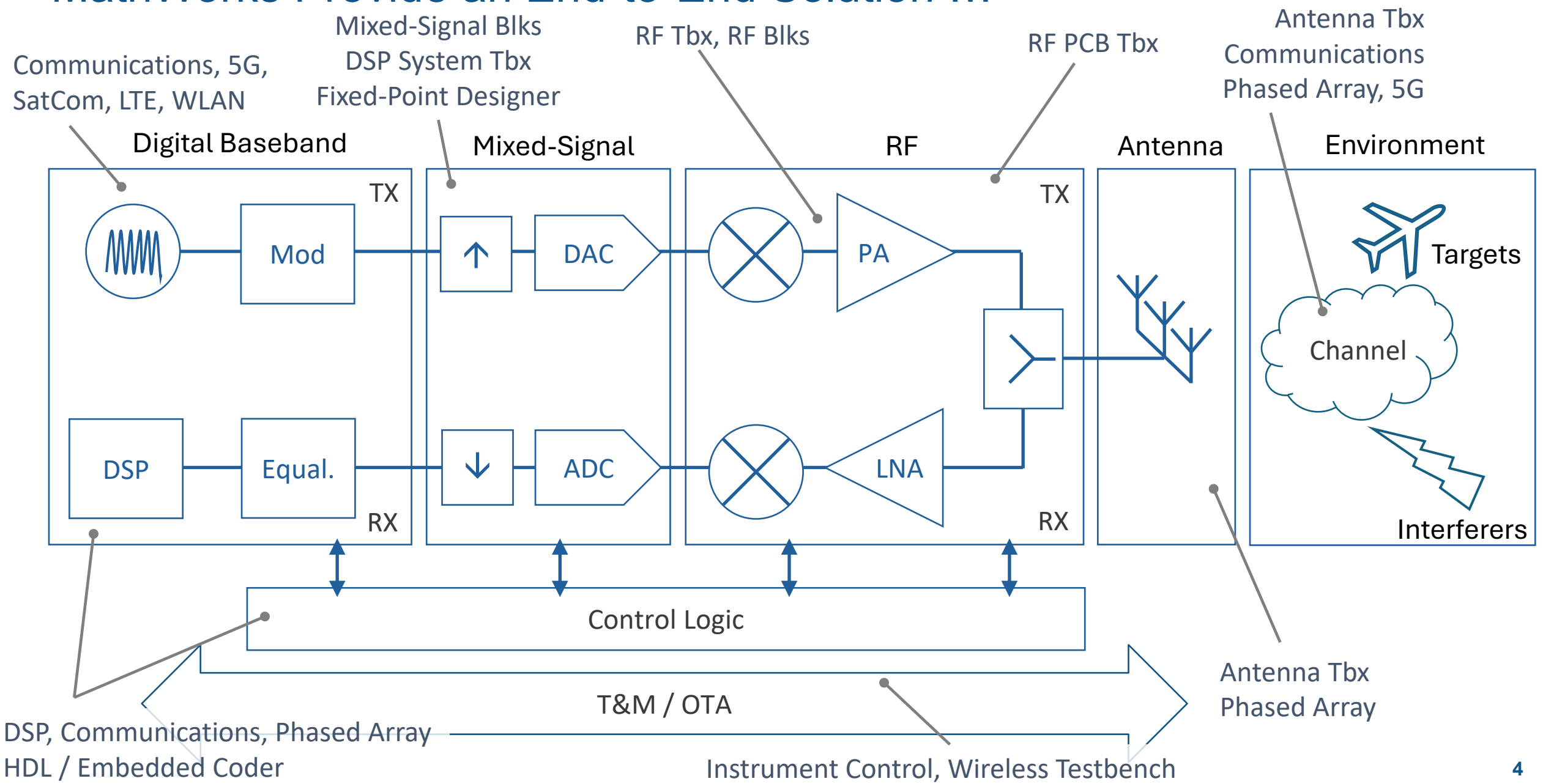


Algorithms Implementation



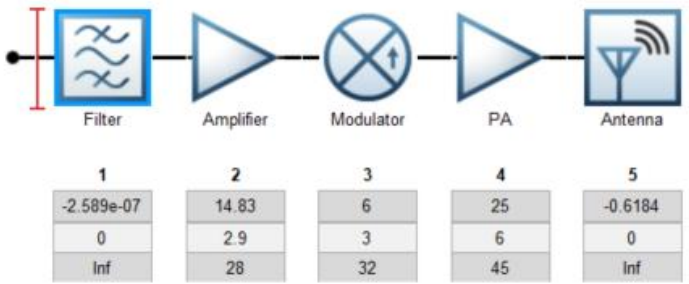
Antenna Arrays

MathWorks Provide an End-to-End Solution …

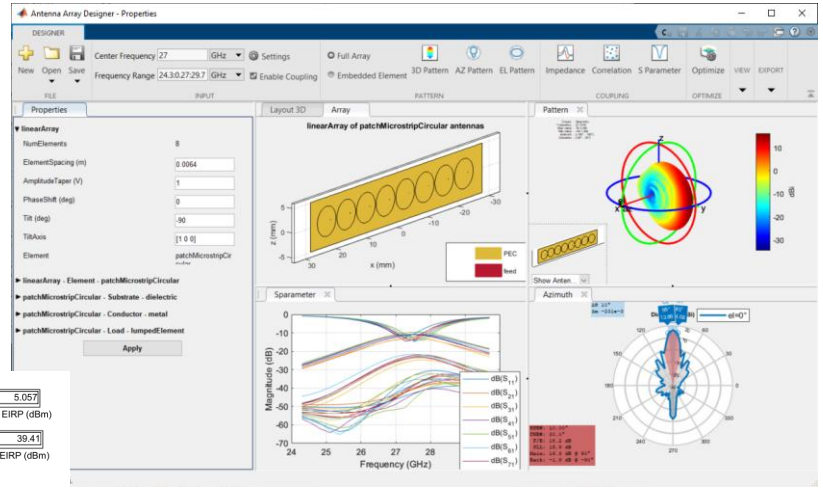


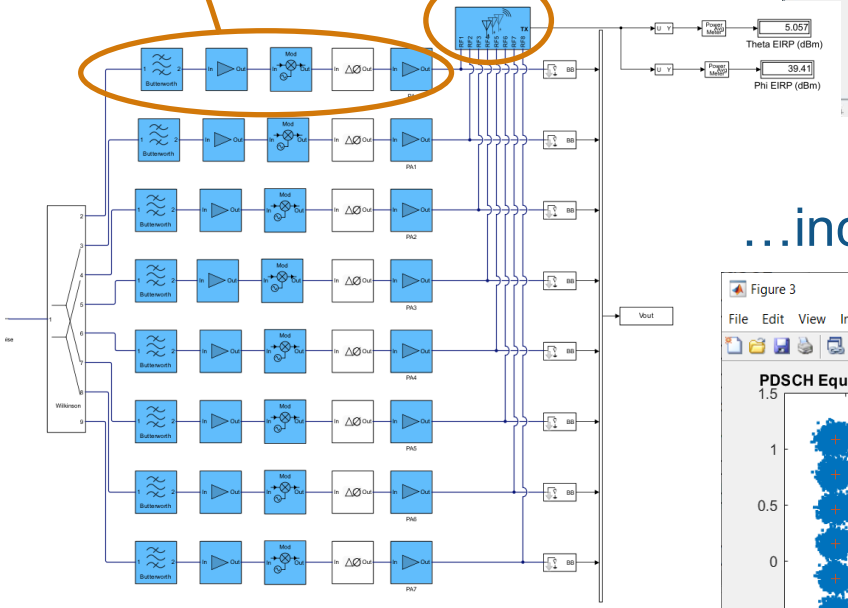
Model Based Design for Antenna2Bits Systems

From single chain budget analysis …



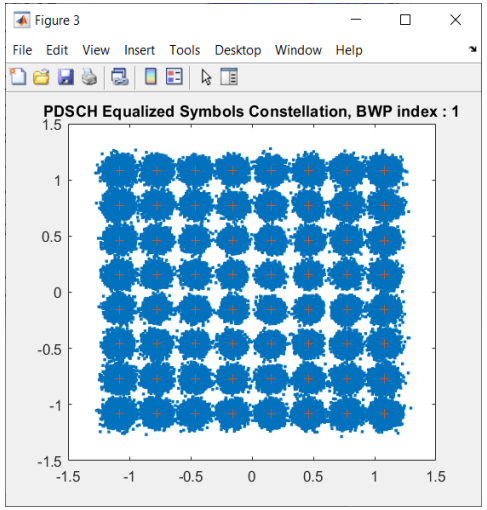
… and antenna array design

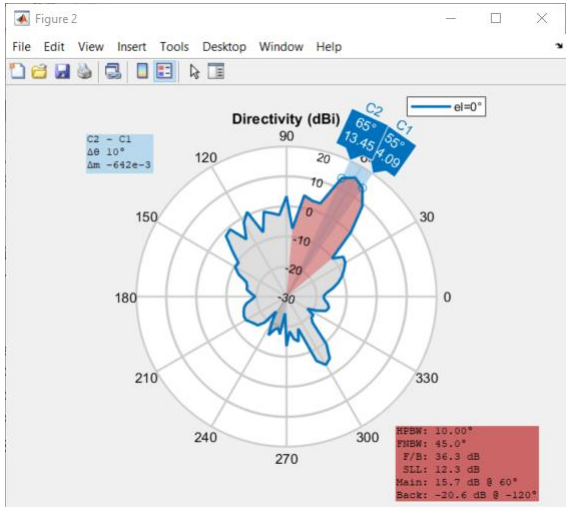




to system integration…

…including baseband and DSP algorithms

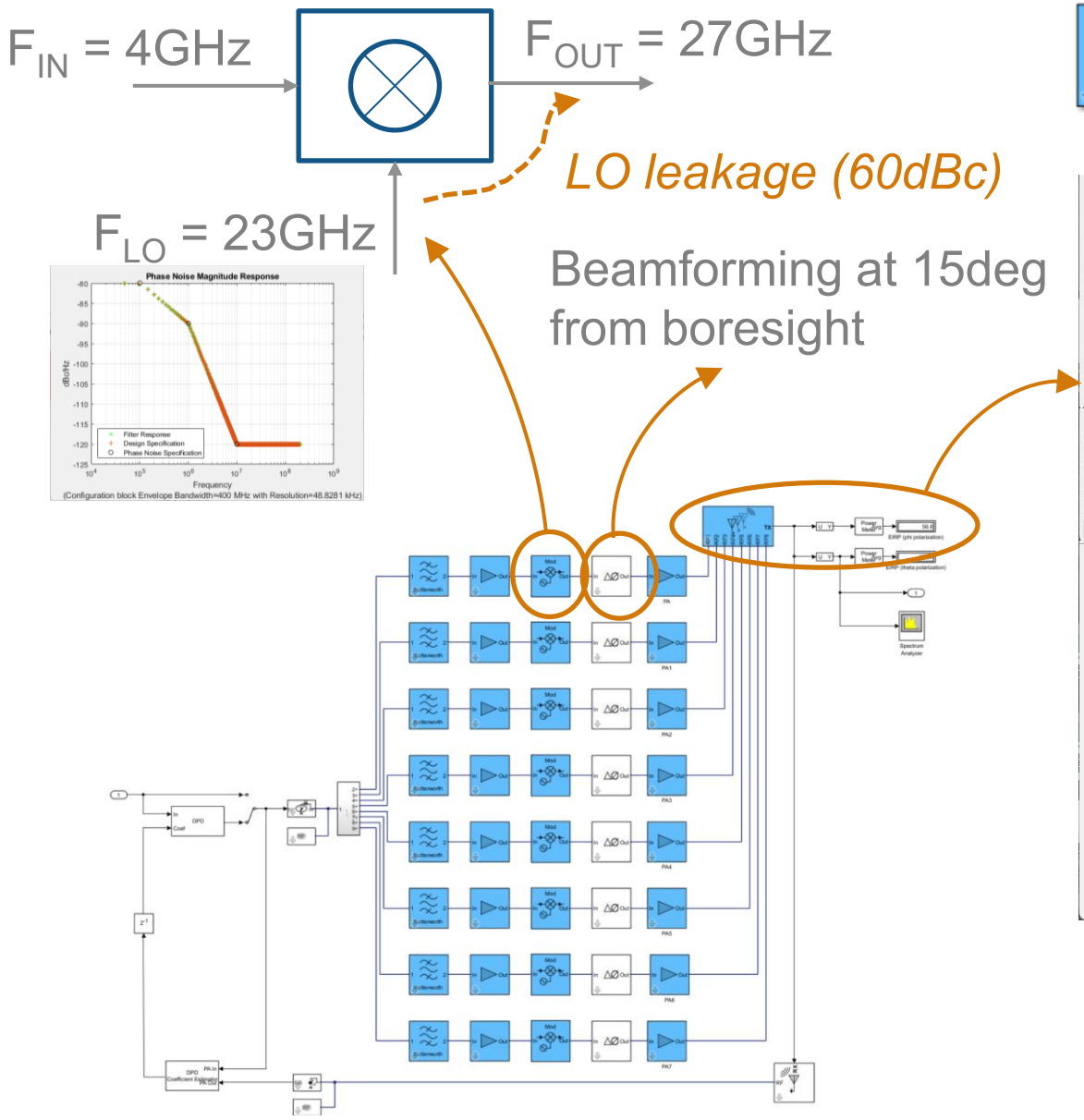


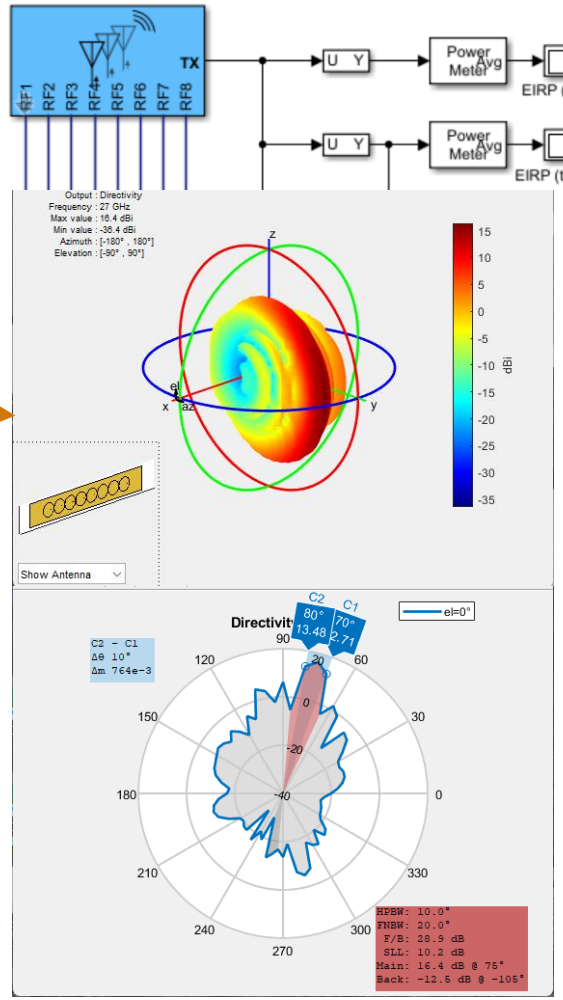


Examples

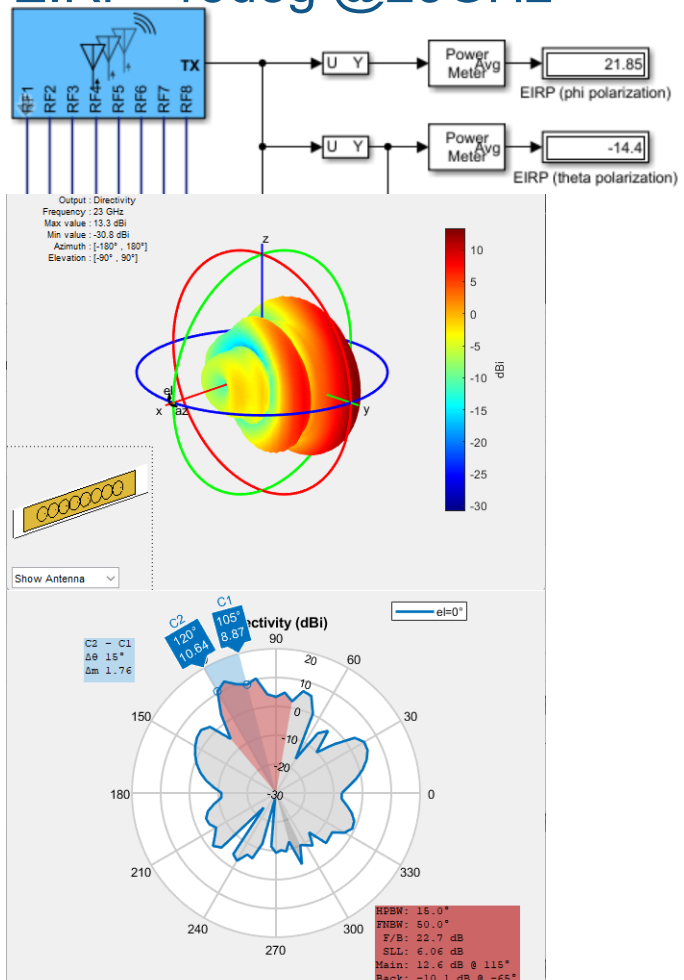
1. Testing transmitter emissions

Estimate the impact of antenna coupling and squinting on beam steering algorithms





EIRP 15deg @27GHz



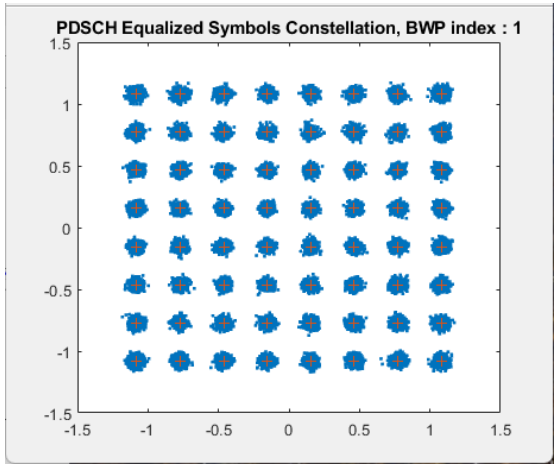
EIRP -15deg @23GHz

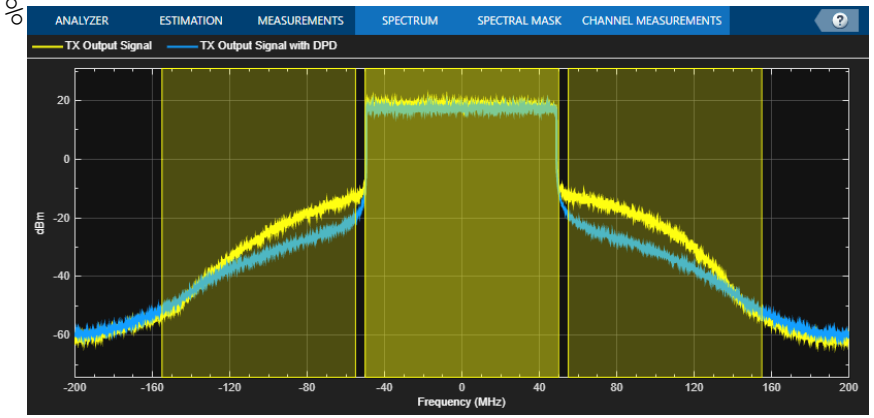
2. Developing and testing transmitter linearization

Use standard-compliant wideband waveforms for dynamic EVM measurements

AMCAD Engineering joins SIMULIA

No DPD EVM RMS = 3.4797%

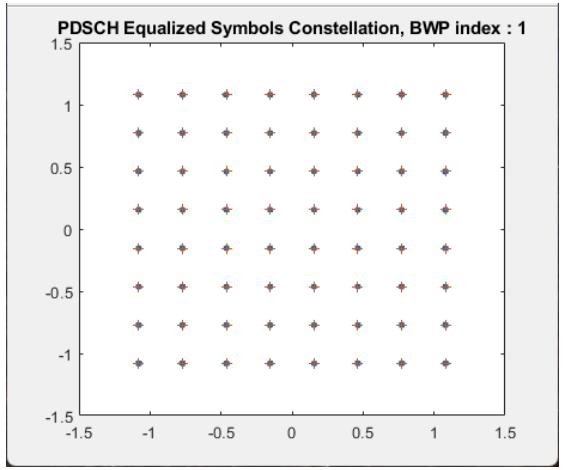


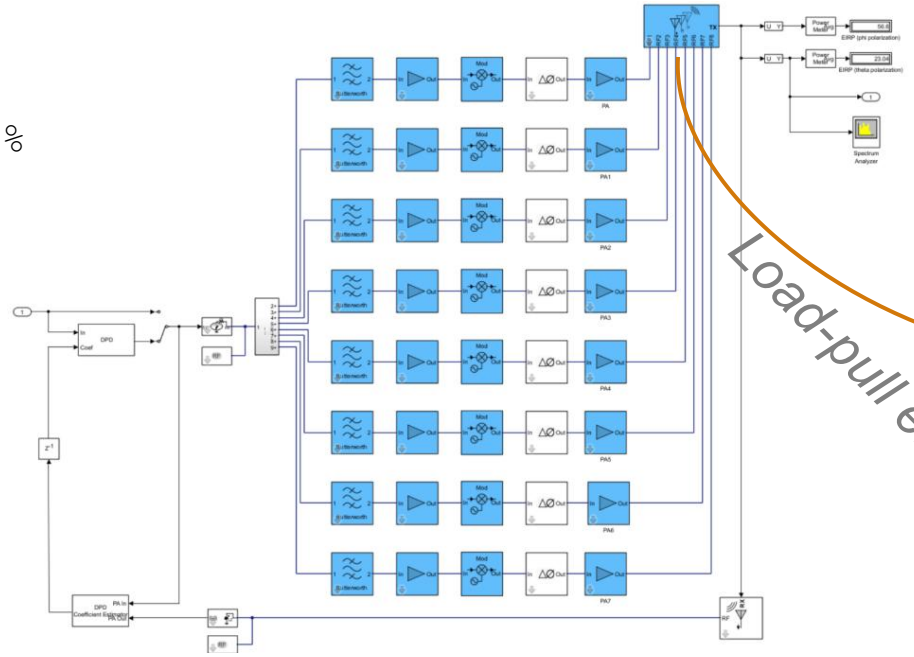


No DPD ACPR = 37dBc

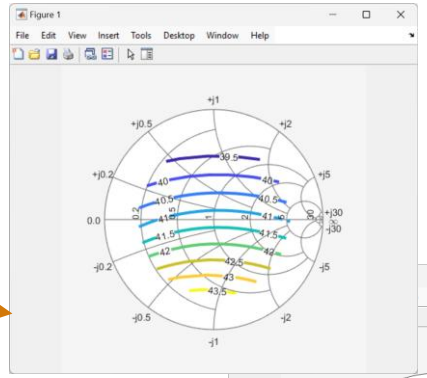
With DPD ACPR = 45dBc

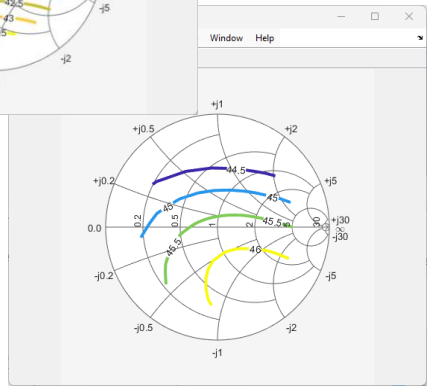
With DPD EVM RMS = 0.44591%





Load-pull effects

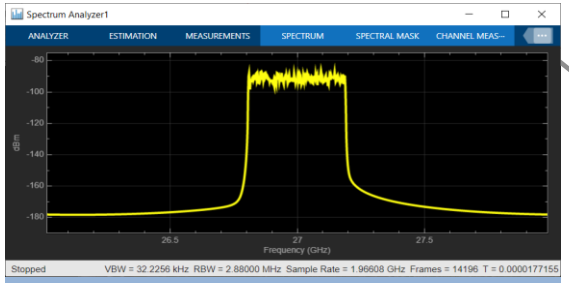




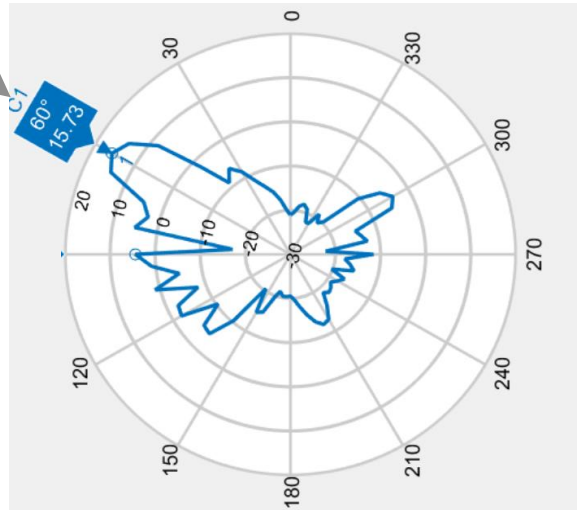
mathworks.com/videos/series/linearization-of-upcoming-high-efficient-rf-power-amplifiers.html

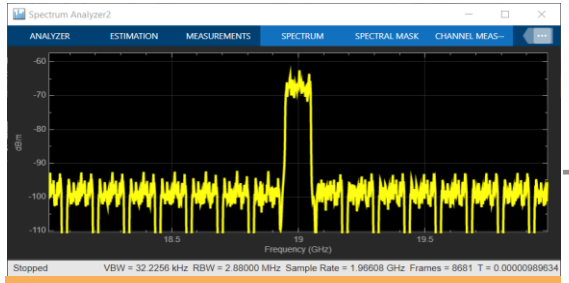
3. Simulate in- and out-of-band interfering signals

Model interference scenarios, sensing, and multi-standard coexistence



3GPP FR2 TM3.1
Desired signal @27GHz
Az El = [60 0]
Power = -70dBm





OFDM 100MHz BW
Interferer @19GHz
(image frequency)
Az El = [40 0]
Power = -70dBm

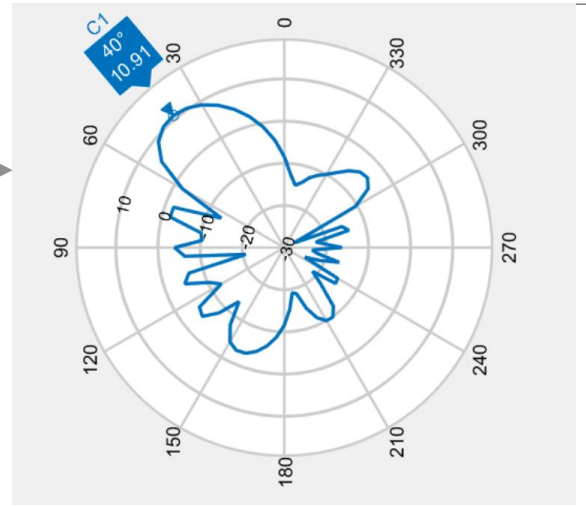
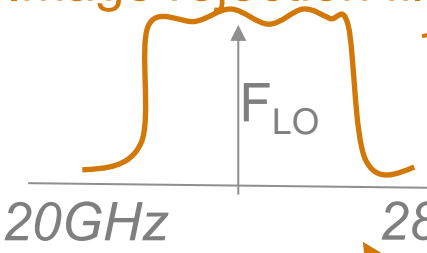


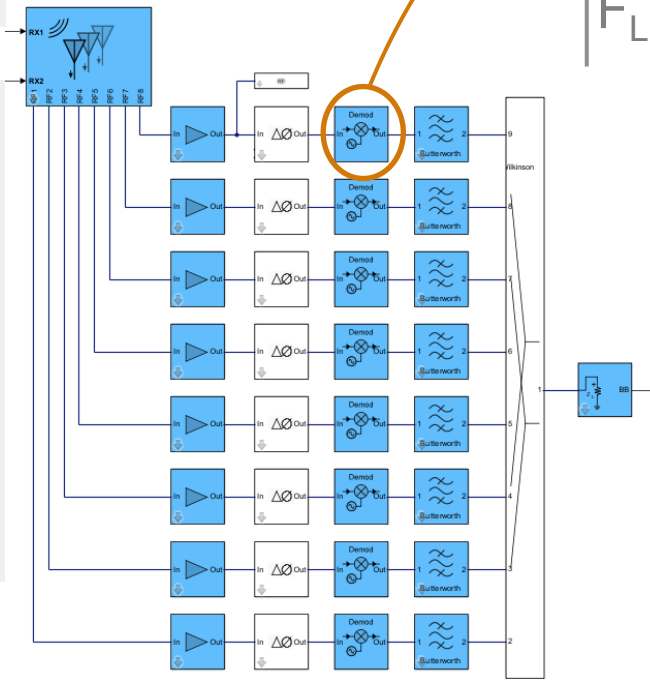
Image rejection filter

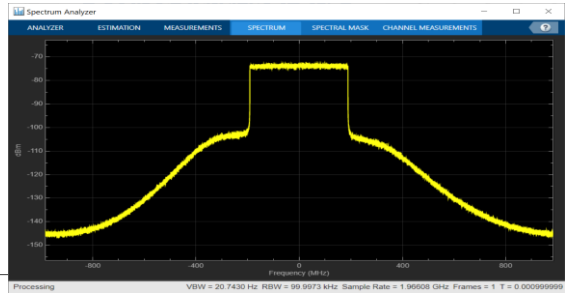


Finite isolation

F_{IN} = 27GHz
F_{INTERF} = 19GHz

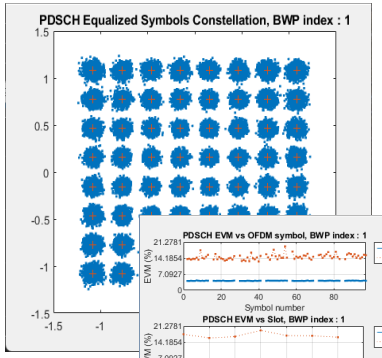
F_{IF_INTERF} = 4GHz
F_{IF} = 4GHz
F_{LO} = 23GHz

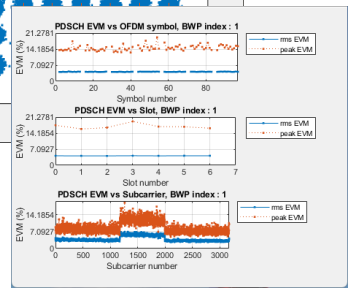




RMS EVM = 3.6%

With interfering signal
RMS EVM = 4.2%

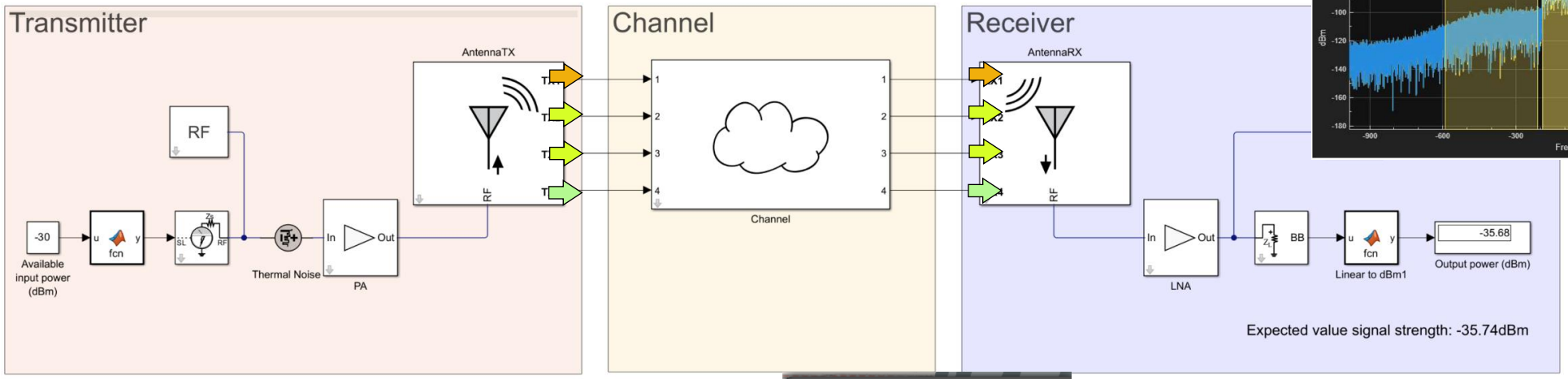


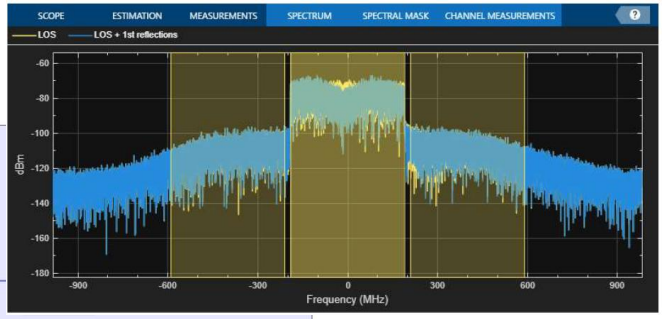


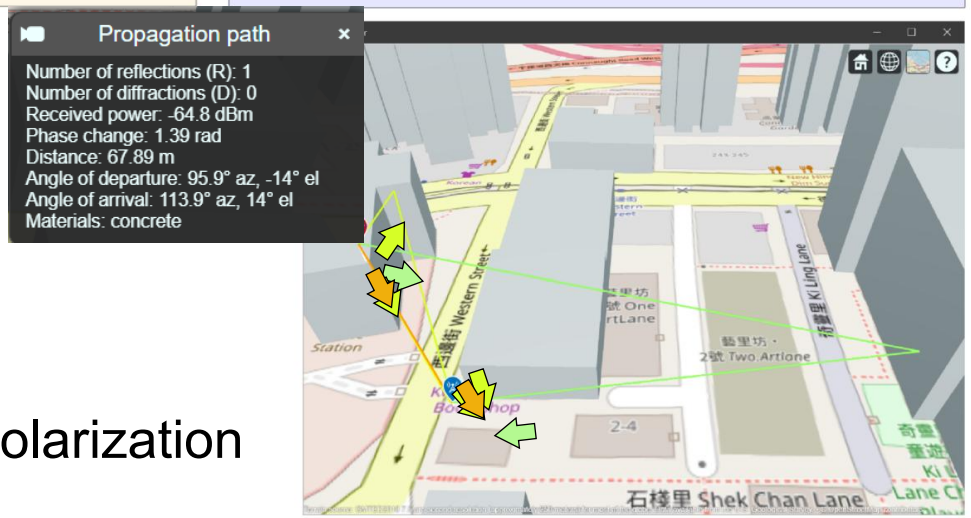
4. Ray tracing channel modeling for wideband signals

Model end-to-end scenarios including dispersion, mismatches, and coupling

- Use ray tracing to account for multiple paths in channel between antennas:







- Scenario modelling including buildings
- Multiple radiated/incident directions
- Path loss and phase shift for each path
- Model coupling in between antenna elements
- Take into account frequency dependency and polarization
- Predict fading effects

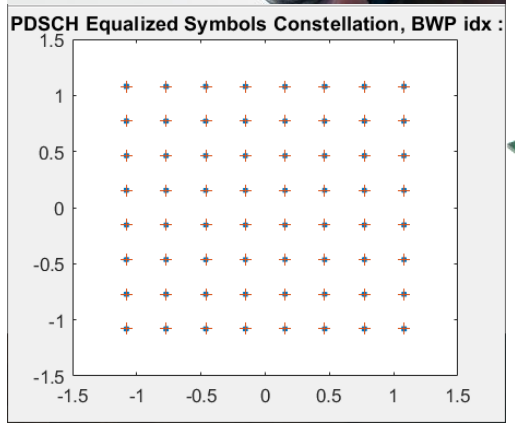
mathworks.com/help/simrf/ref/channel.html

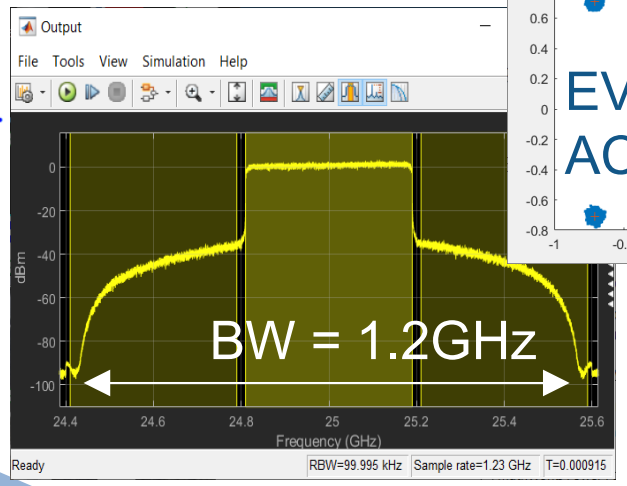
Transmitter scenario modeling - 3GPP 5GNR FR2 waveforms

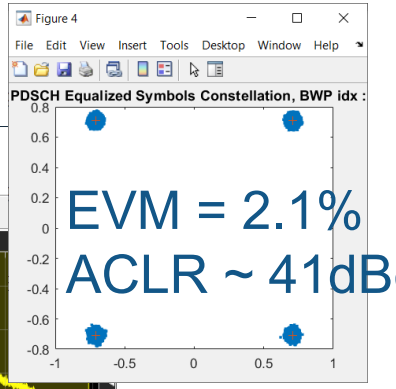
```
%% Setup for ACLR measurements
rc = "NR-FR2-TM1.1"; % Reference channel(NR-TM or FRC)
bw = "400MHz"; % Channel bandwidth
scs = "120kHz"; % Subcarrier spacing
dm = "TDD"; % Duplexing mode
tmwavegen= hNRReferenceWaveformGenerator(rc,bw,scs,dm);
[txWaveform,tmwaveinfo,resourcesinfo] = generateWaveform(tmwavegen,..
    tmwavegen.Config.NumSubframes);
```

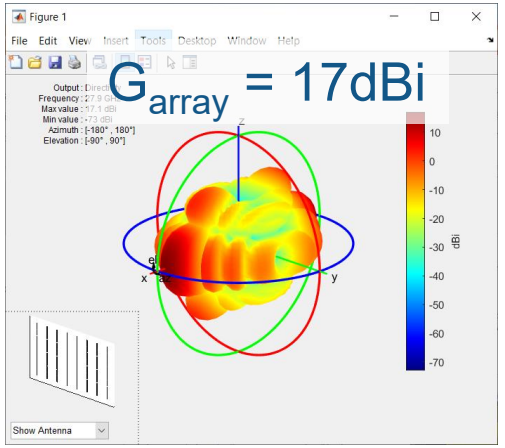


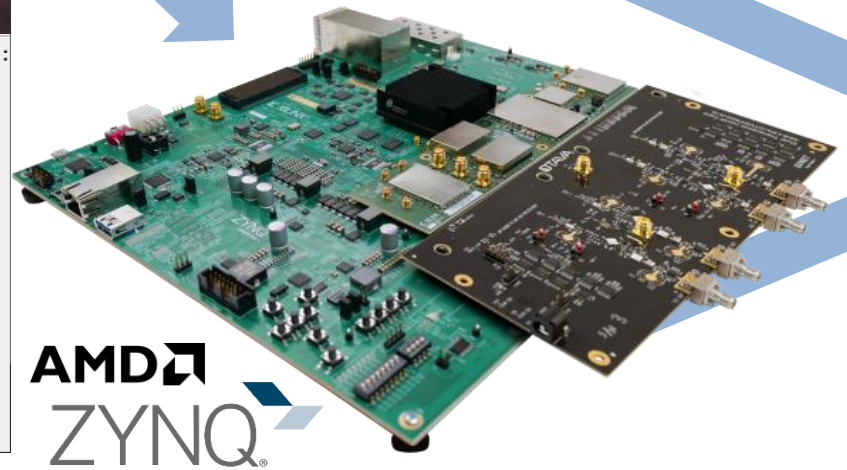
Reference waveform
FR2 TM1.1 400MHz

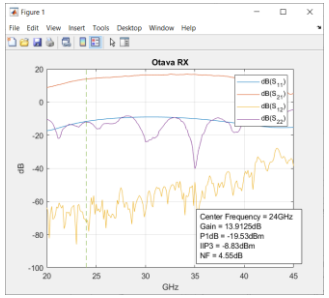


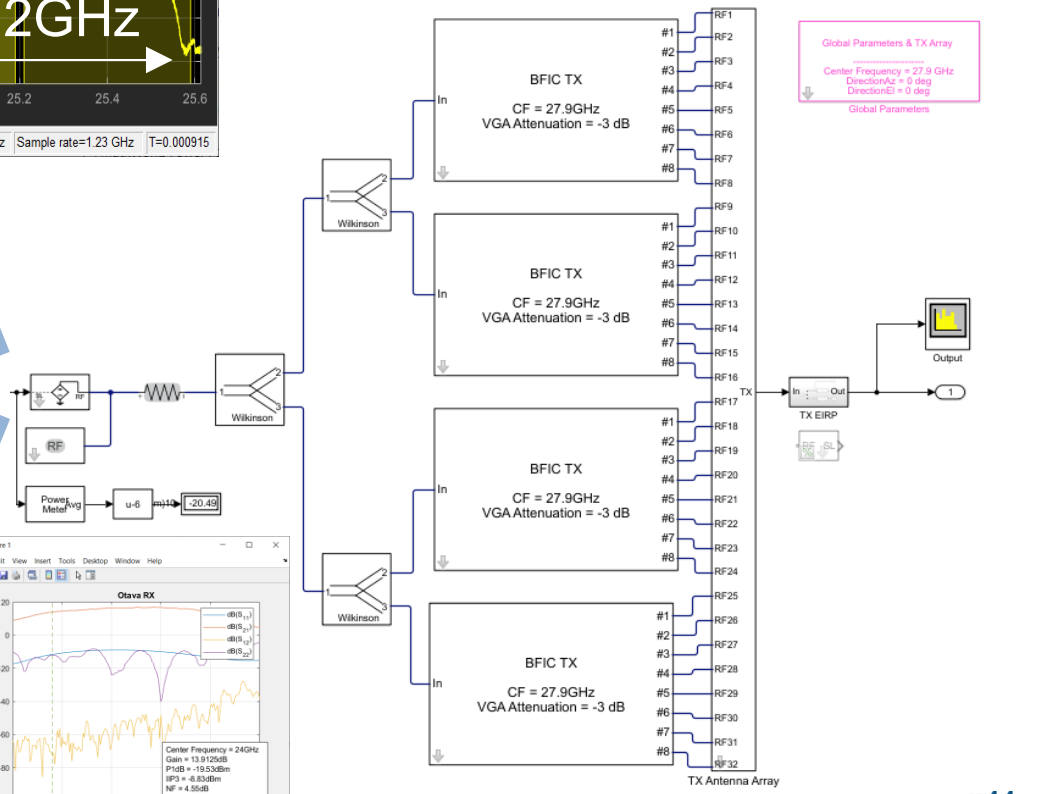








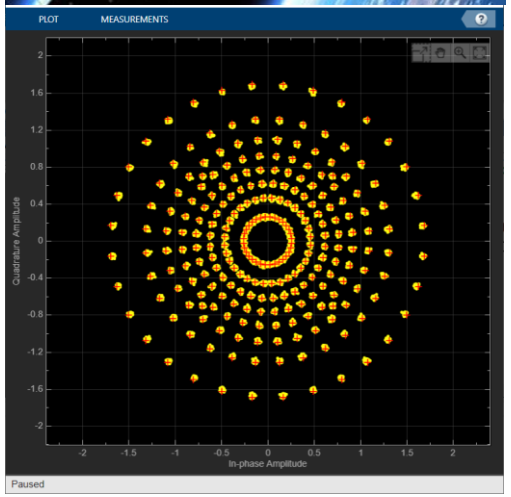




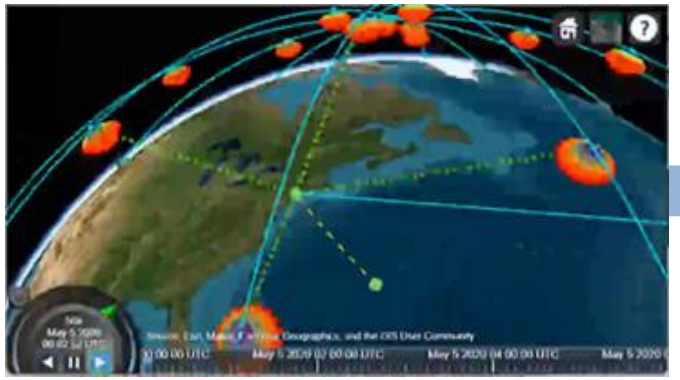
Receiver scenario modeling: SatCom DVB-S2X waveform

Reference waveform



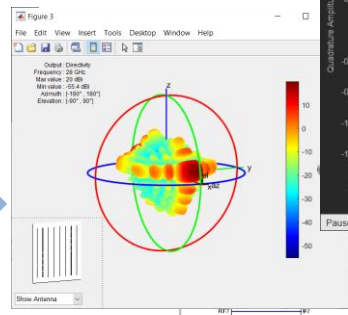


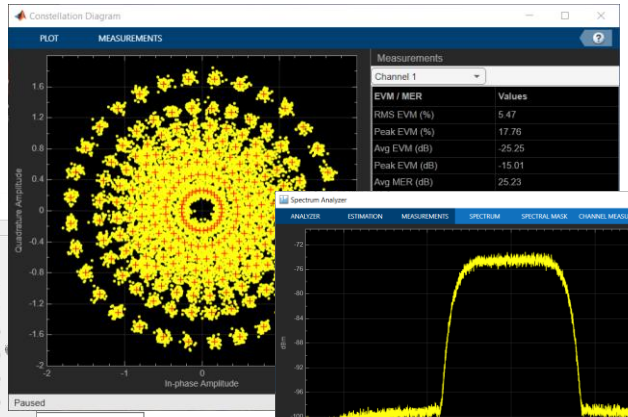
NTN Channel model

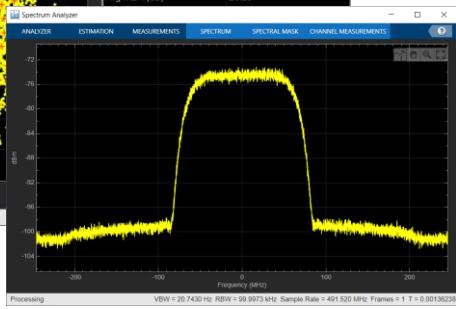


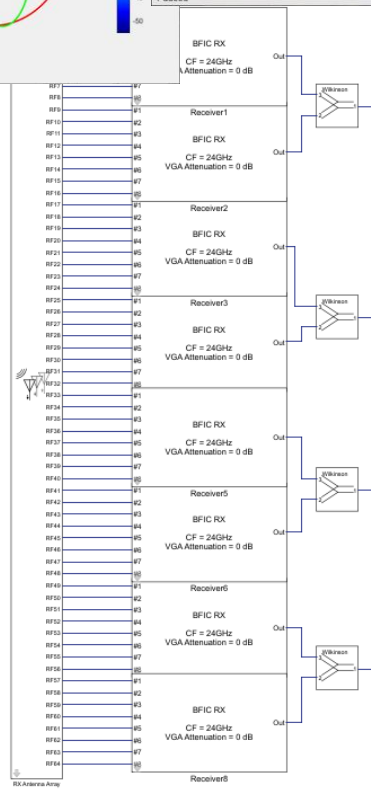
```
commonParams.CarrierFrequency = 24e9; %Hz
commonParams.ElevationAngle = 55; %deg
commonParams.SatelliteAltitude = 250000; %m
commonParams.SatelliteSpeed = 7500; %m/s
commonParams.MobileSpeed = 0; %m/s
commonParams.SampleRate = Fsamp; %Hz
commonParams.NumSinusoids = 48;
ntnNarrowbandParams = commonParams;
ntnNarrowbandParams.NTNChannelType = "Narrowband";
ntnNarrowbandParams.Environment = "Suburban";
ntnNarrowbandParams.AzimuthOrientation = 0;
ntnNarrowbandParams.FadingTechnique = "Sum of sinusoids";
```

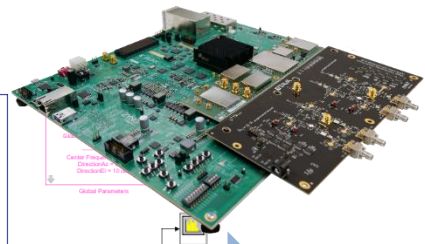
BFIC RX









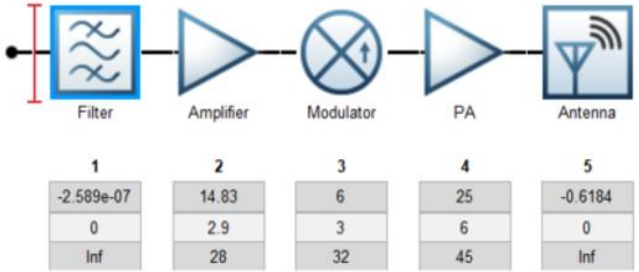


Avg P_{out} ~ -43.7dBm
$PAPR_{out}$ = 7.9dB
SNR_{out} = 25dB
EVM_{out} = 5.5%

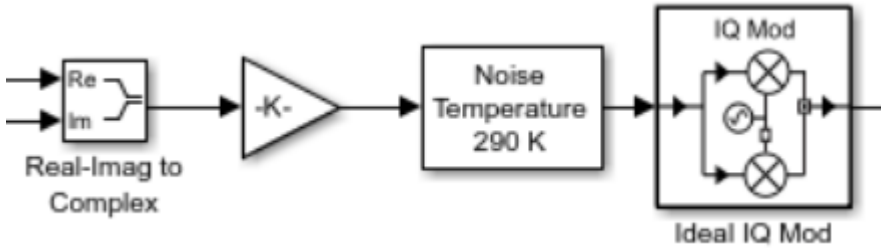
Part 3A — RF Workflow

RF Workflow

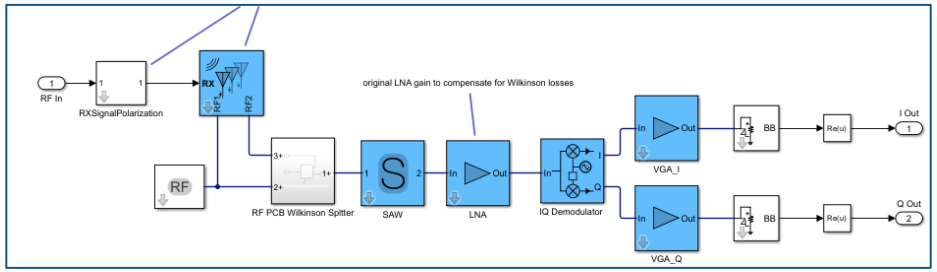
RF Budget Analysis



Idealized Baseband



Circuit Envelope



Idealized Baseband

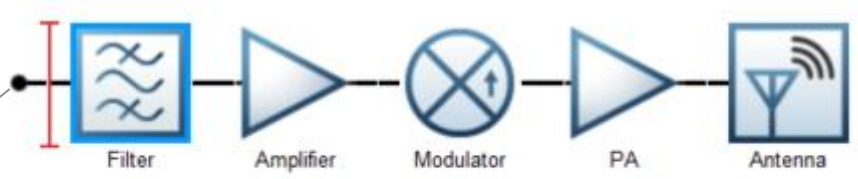
```
RFmixer = rf.Mixer('Model','iqmod');
RFfilter = rf.Filter('ResponseType','Bandpass', ...
    'UseFilterOrder',false, ...
    'PassFreq_bp',fc+[-(sr/2-(sr/2*0.6)) sr/2-(sr/2*0.6)], ..
    'PassAtten',0.7,...
    'StopFreq_bp',fc+[-(sr/2-(sr/2*0.5)) sr/2-(sr/2*0.5)], ..
    'StopAtten',60, ...
    'SampleRate',sr, ...
    'RF',fc);
RFHPA = rf.Amplifier('Gain',gain);
```

Integrate with Baseband

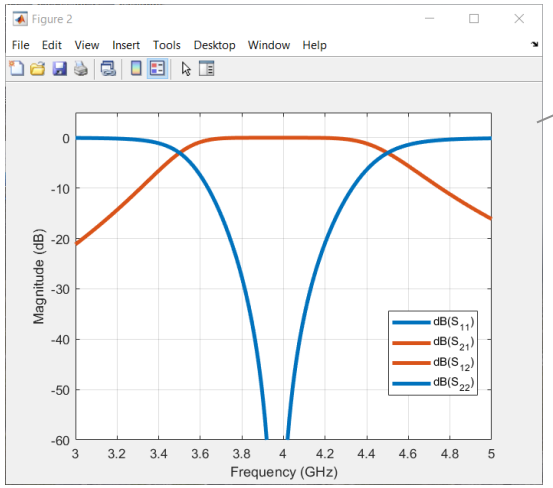
RF Wideband System Design Based on Specs and Measurement Data

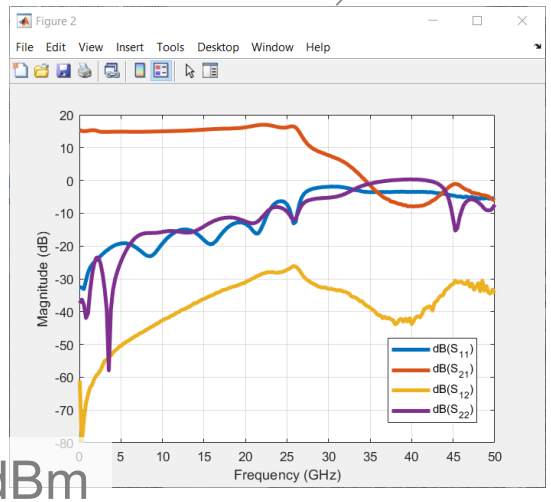
Perform budget analysis over frequency: account for dispersion, losses, impedance mismatches

Pin = -30dBm
Input Frequency = 4GHz



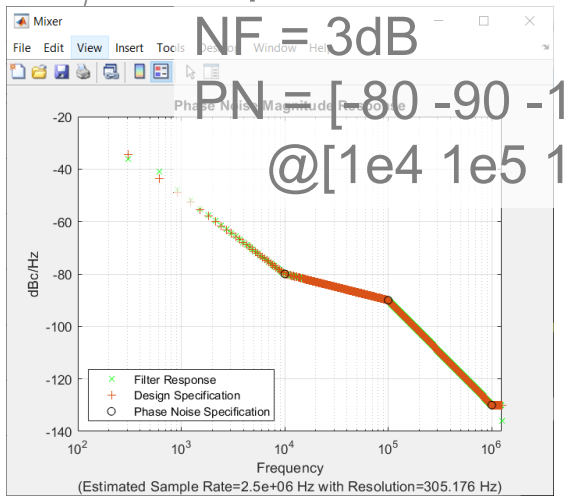
1	2	3	4	5
-2.589e-07	14.83	6	25	-0.6184
0	2.9	3	6	0
Inf	28	32	45	Inf

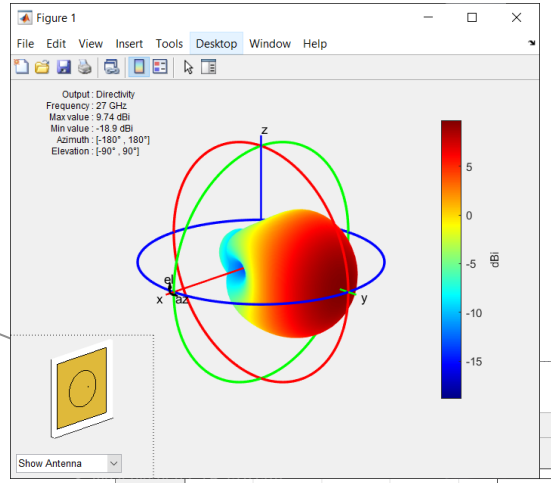


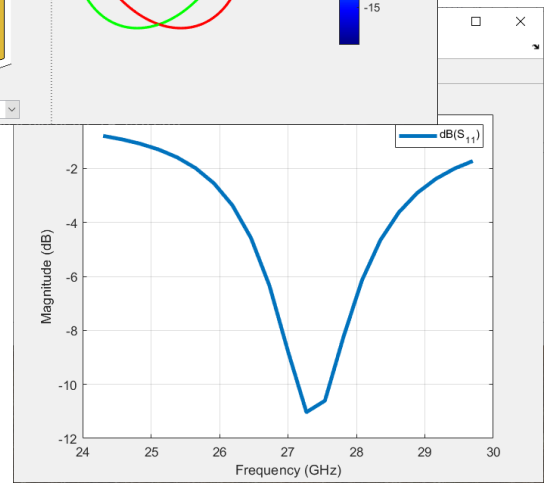


Oip3 = 28dBm
NF = 2.9dB

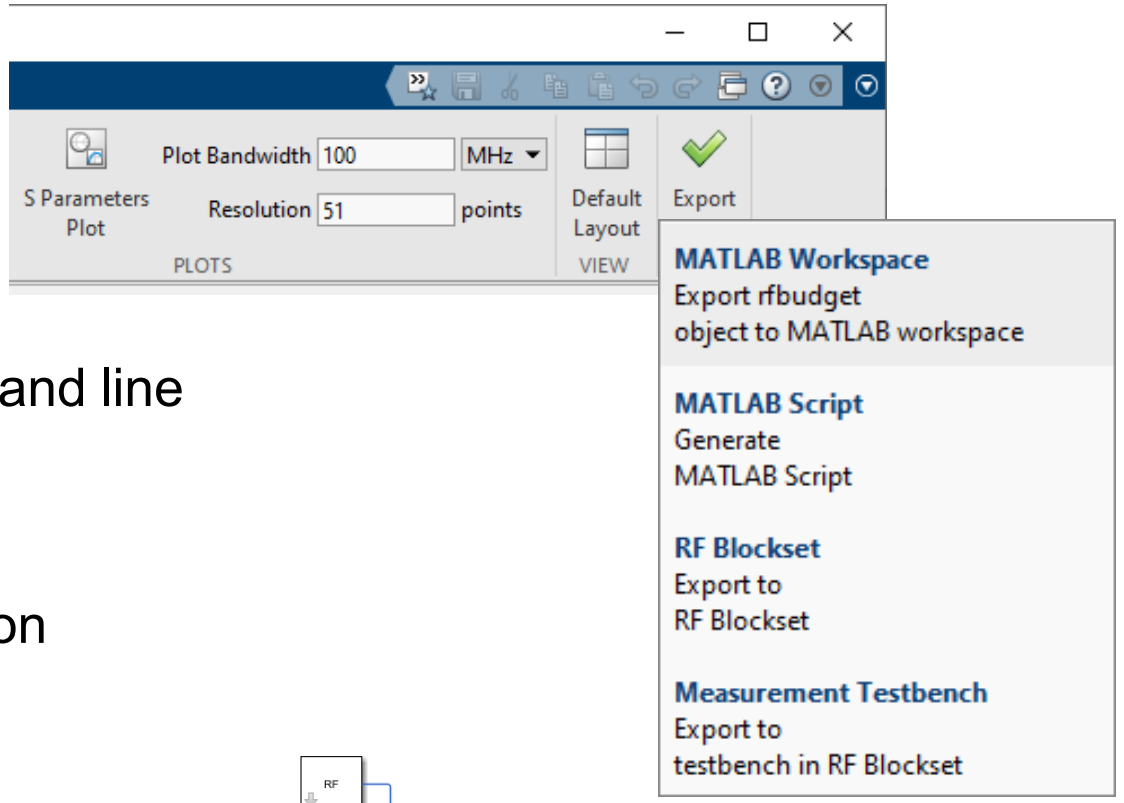
LO frequency = 23GHz
Oip3 = 32dBm
NF = 3dB
PN = [-80 -90 -130]dBc/Hz
@[1e4 1e5 1e6]Hz



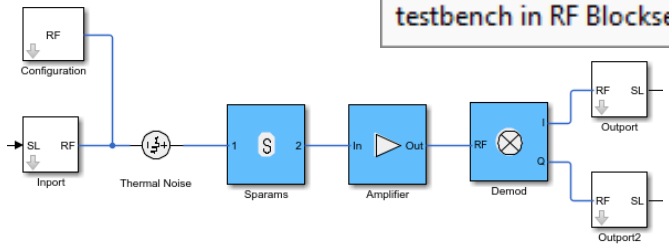




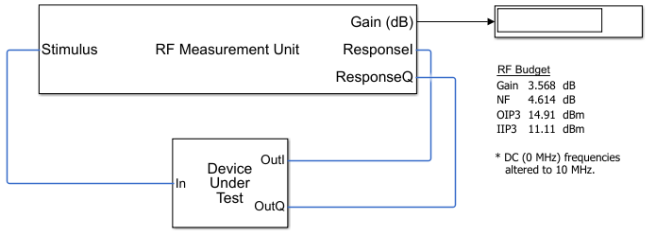
Export from RF Budget Analyzer



- Export to MATLAB workspace object
  - Bidirectional workflow between app and command line
- Export to MATLAB script
  - Automate corner analysis and chain optimization
- Export to RF Blockset
  - Generate model for HB and Circuit Envelope simulation
  - Automatic configuration of Input/output ports

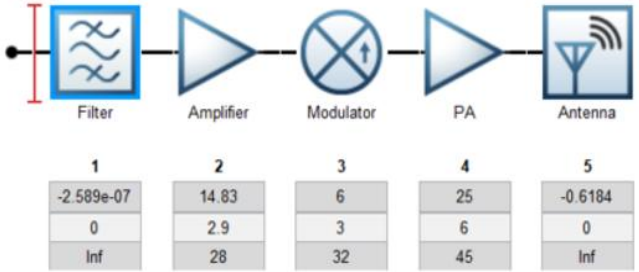


- Export to measurement testbench
  - Measure gain, noise, IP3, IP2, DC offset, and image rejection
  - Validate that the RF system with Circuit Envelope simulation



RF Workflow

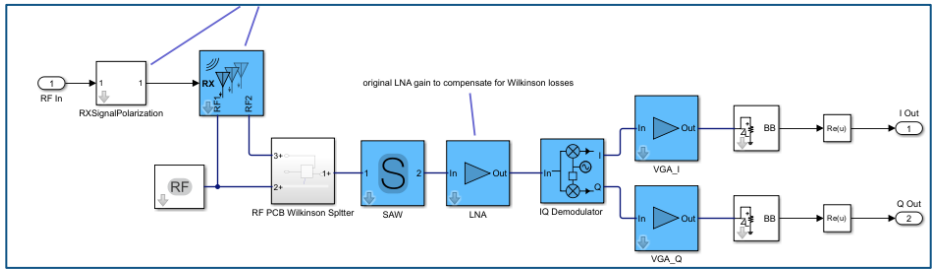
RF Budget Analysis



Idealized Baseband



Circuit Envelope



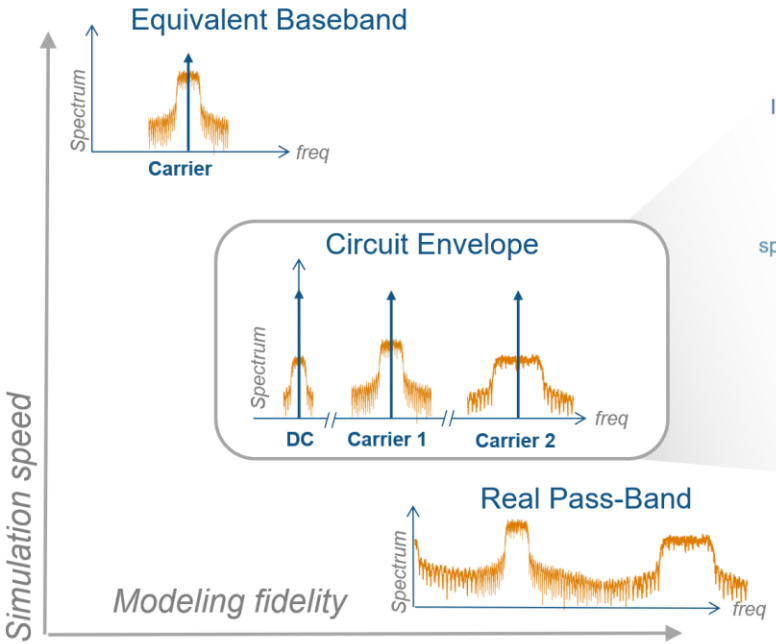
Idealized Baseband

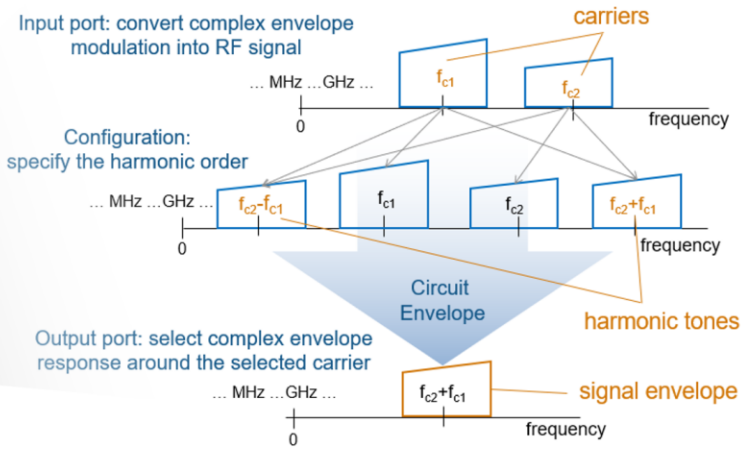


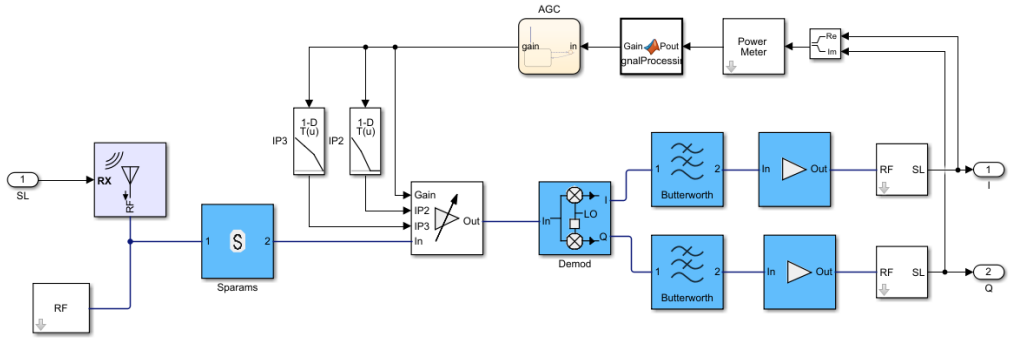
Integrate with Baseband

RF System Simulation with Circuit Envelope

- Design the architecture and define the specs of the RF components at the system-level
- Perform fast RF simulation with multicarrier circuit envelope
- Integrate RF front ends with adaptive algorithms such as DPD, AGC, beamforming
- Test and debug the implementation of the transceiver before going in the lab
- Use models and measured data to gain insights in your design

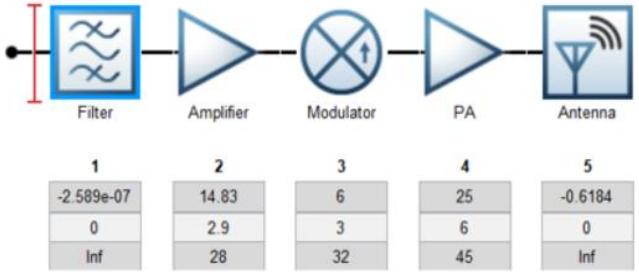






RF Workflow

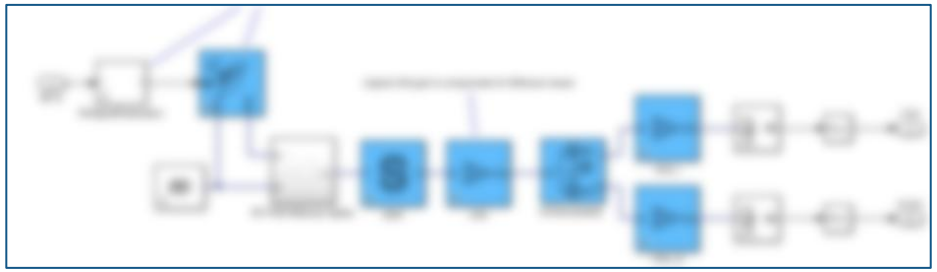
RF Budget Analysis



Idealized Baseband



Circuit Envelope



Integrate with Baseband

Idealized Baseband

```
RFmixer = rf.Mixer('Model','iqmod');
RFfilter = rf.Filter('ResponseType','Bandpass', ...
    'UseFilterOrder',false, ...
    'PassFreq_bp',fc+[-(sr/2-(sr/2*0.6)) sr/2-(sr/2*0.6)], ..
    'PassAtten',0.7,...
    'StopFreq_bp',fc+[-(sr/2-(sr/2*0.5)) sr/2-(sr/2*0.5)], ..
    'StopAtten',60, ...
    'SampleRate',sr, ...
    'RF',fc);
RFHPA = rf.Amplifier('Gain',gain);
```

Idealized Baseband Library: Simplified Modeling of RF Components

- Model the chain of RF components with datasheet specifications
- Complex equivalent baseband approach: in-band odd-order non-linearity
- Same models in MATLAB and Simulink: unidirectional models (no impedance mismatches)
- Support frame-based processing and C code generation

Objects

- `rf.PAmemory`
- `rf.Amplifier`
- `rf.Mixer`
- `rf.Sparameter`
- `rf.Filter`

`rf.Amplifier`

- Noise Figure
- IP3, Psat, P1dB AM-AM/AM-PM

`rf.Mixer`

- Phase Noise
- I/Q gain-phase Imbalance

`rf.Sparameter`

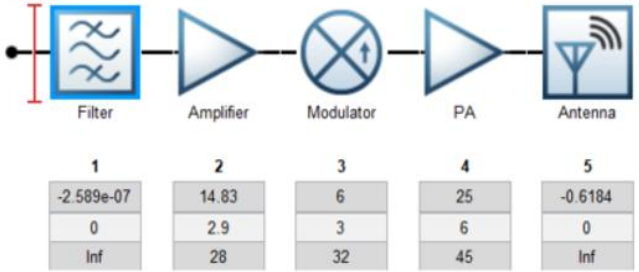
- Touchstone files (dispersion)

`rf.PAmemory`

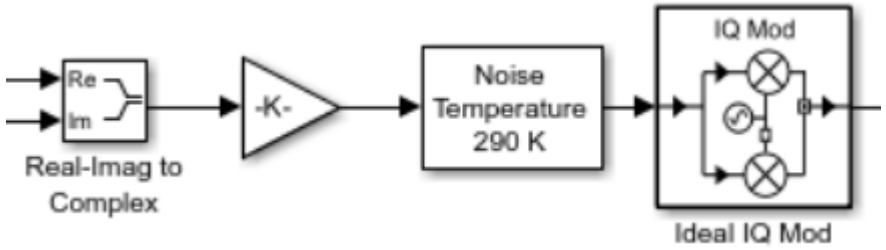
- Nonlinearity Memory

RF Workflow

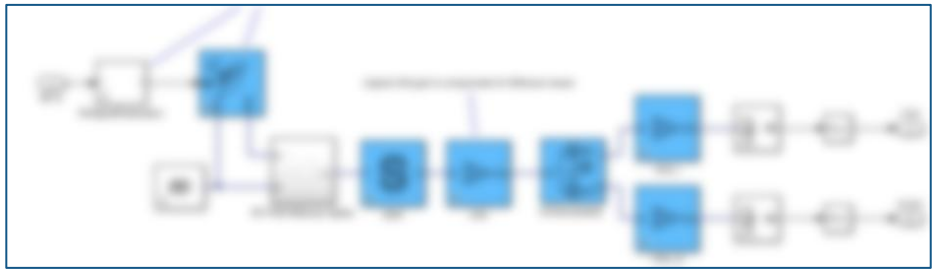
RF Budget Analysis



Idealized Baseband



Circuit Envelope

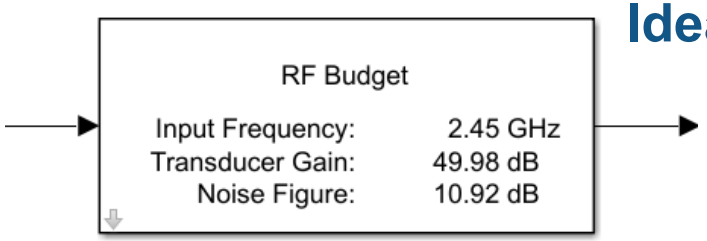


Integrate with Baseband

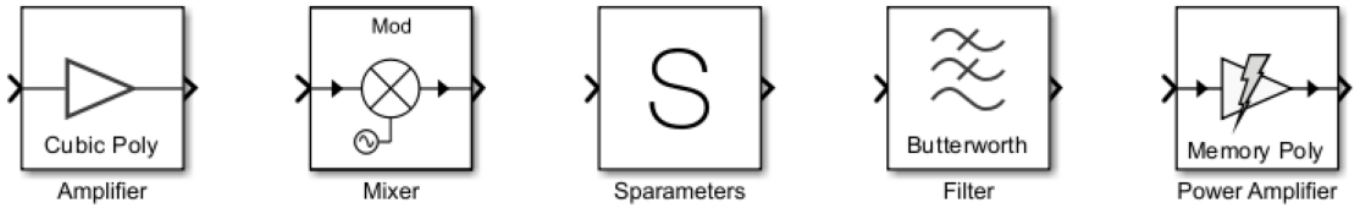
Idealized Baseband



Idealized Baseband: Models

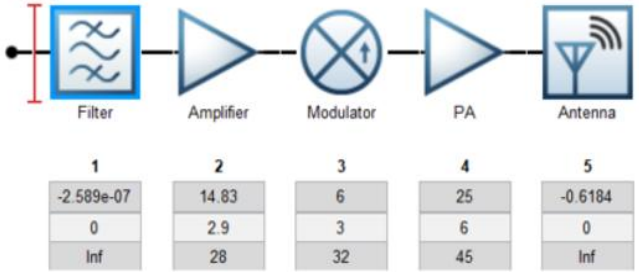


- Amplifier:
  - Odd-order nonlinearity and saturation
  - AM-AM/AM-PM
- Mixer:
  - Modulator, demodulator, also quadrature models
  - 2-port model including LO signal with arbitrary phase noise profile
- S-parameters:
  - Simulate only transmission (S_{21})
  - Use rational fitting (time-domain) or convolution (frequency domain)
- Filter:
  - Ideal transfer function: Butterworth, Chebyshev, inverse Chebyshev
- Power Amplifier:
  - Generalized memory polynomial

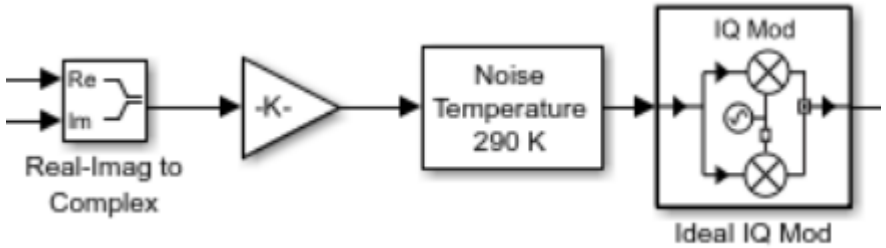


RF Workflow

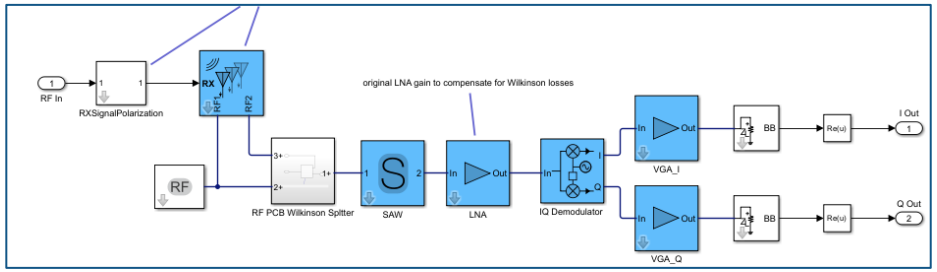
RF Budget Analysis



Idealized Baseband



Circuit Envelope



Idealized Baseband

```
RFmixer = rf.Mixer('Model','iqmod');
RFfilter = rf.Filter('ResponseType','Bandpass', ...
    'UseFilterOrder',false, ...
    'PassFreq_bp',fc+[-(sr/2-(sr/2*0.6)) sr/2-(sr/2*0.6)], ..
    'PassAtten',0.7,...
    'StopFreq_bp',fc+[-(sr/2-(sr/2*0.5)) sr/2-(sr/2*0.5)], ..
    'StopAtten',60, ...
    'SampleRate',sr, ...
    'RF',fc);
RFHPA = rf.Amplifier('Gain',gain);
```

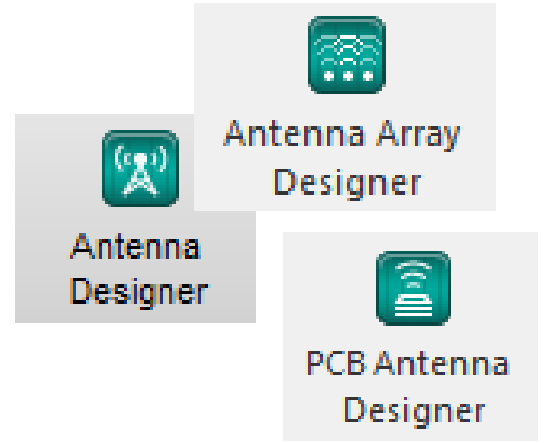
Integrate with Baseband

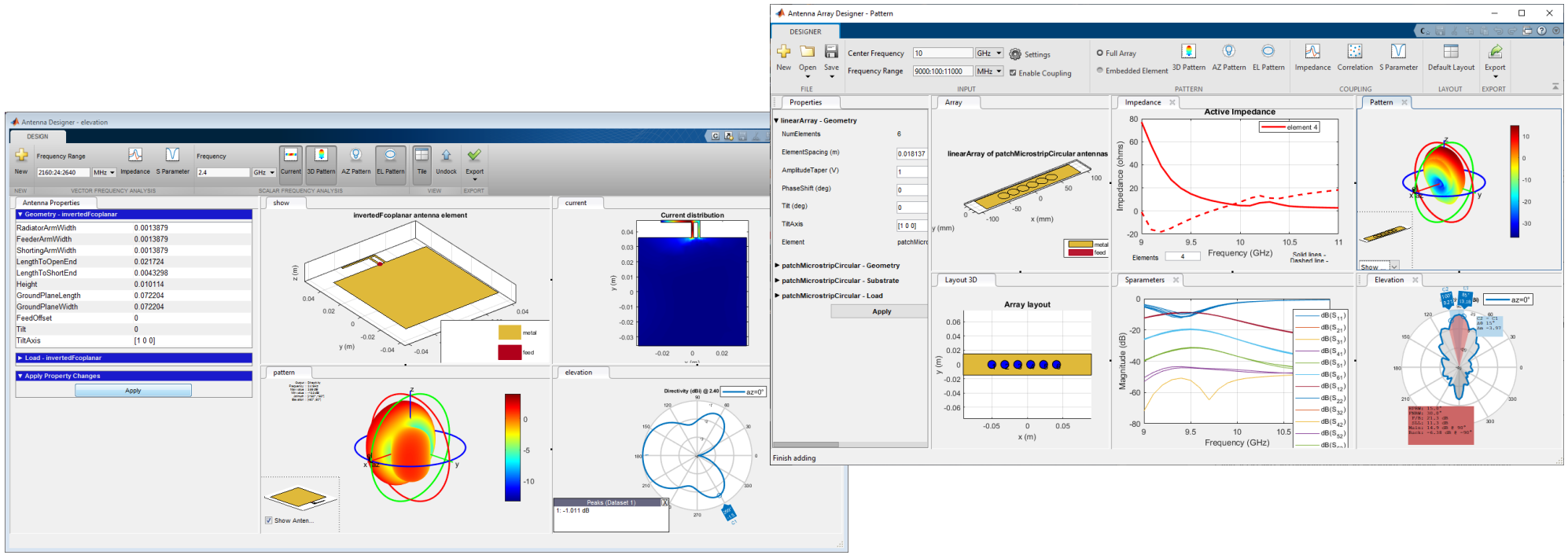
Antenna and Array Design using Electromagnetic Analysis

Antenna Design – Where To Start?

Antenna Designer / Array Designer / PCB Antenna Designer Apps

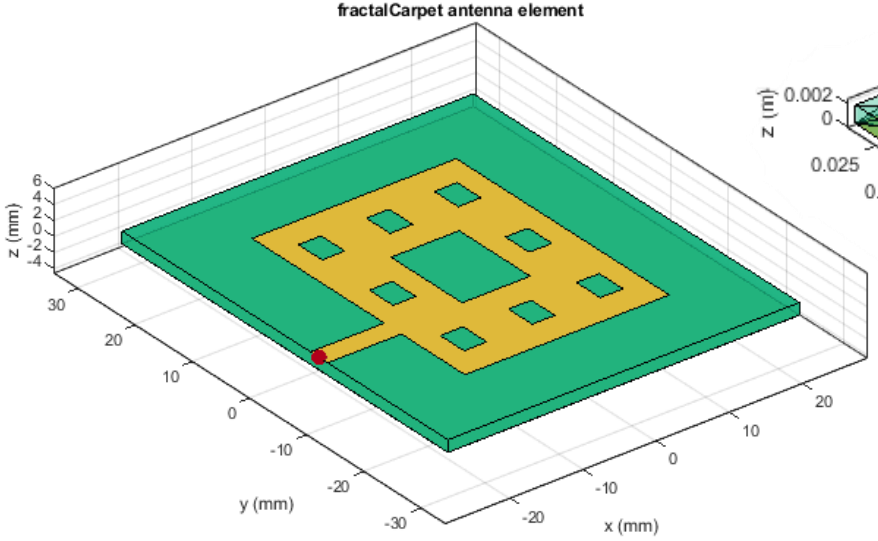
- Select an antenna/array based on the desired specifications
- Design the antenna at the operating frequency
- Visualize results and iterate on antenna geometrical properties
- Generate MATLAB scripts for automation

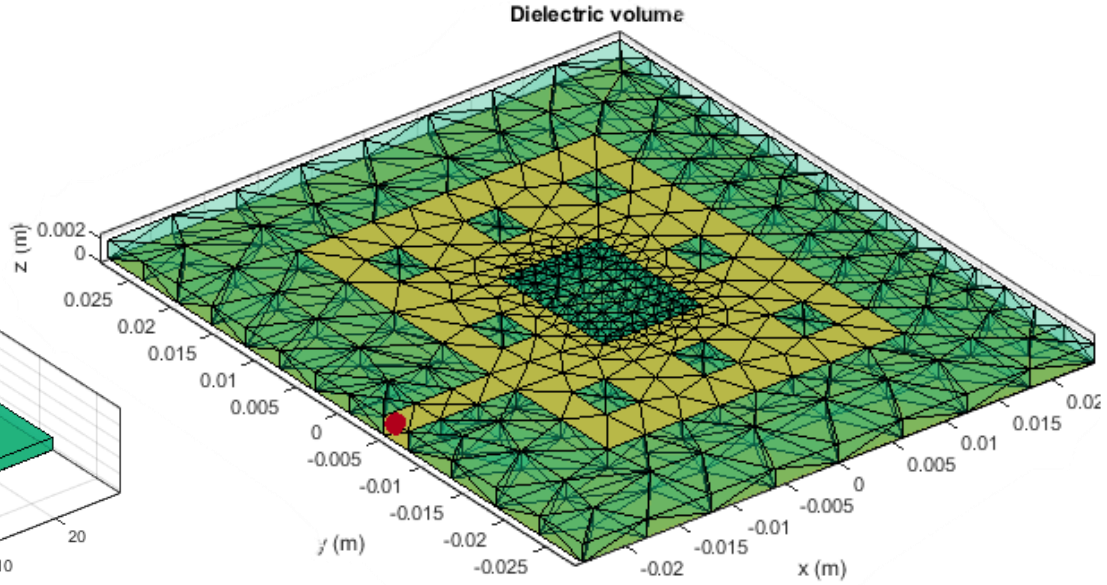


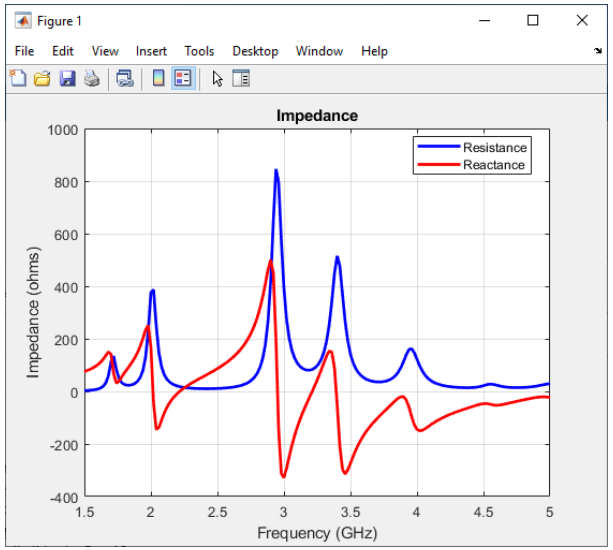


Antenna Electromagnetic Analysis

- Full-wave solver: Method of Moments
- Analysis-driven automatic mesh generation: triangles for surface, tetrahedra for volume
  - Manual control on meshing for advanced use cases
- Validated against state-of-the-art results

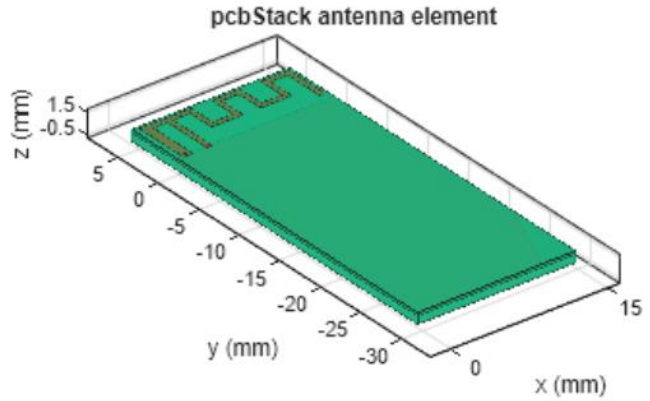


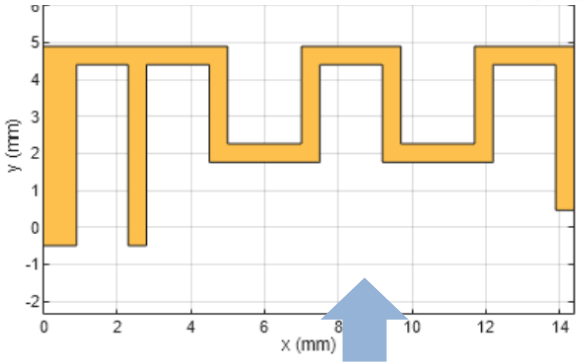




Programmatic workflows unlock what GUI-based tools cannot

- Code is AI-readable. GUIs are not.
- Every dimension is a variable. Instant parameterization.
- GenAI as your antenna design co-pilot.
- Closed-loop automation.
- Reproducible, shareable, version-controlled.





MCP Client

MCP

MCP Server

```
ant = pcbStack;
boardshape = antenna.Rectangle(Length=LBS, Width=WBS+Wgp, Center=[LBS/2-0.5e-3,(WBS-Wgp)/2]);
ant.BoardShape = boardshape;
ant.BoardThickness = d.Thickness + d1.Thickness;
gnd = antenna.Rectangle(Length=Lgp, Width=Wgp, Center=[Lgp/2-0.5e-3,-Wgp/2]);
ant.Layers = {Newobj,d,gnd,d1};
ant.FeedDiameter = W2/2;
ant.ViaDiameter = W1/4;
ant.FeedLocations = [W1+1.4e-3+W2/2,-D4/2,3,1];
ant.ViaLocations = [W1/2,-D4/2,3,1];
```

Dielectric and Metal Material Properties

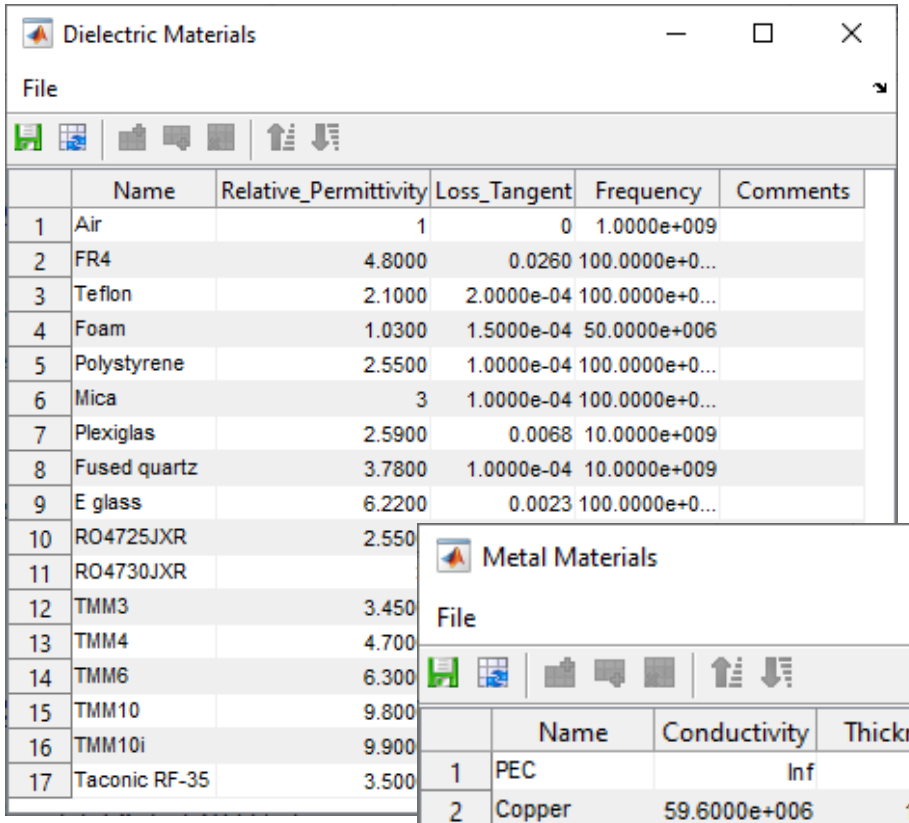
Dielectric materials:

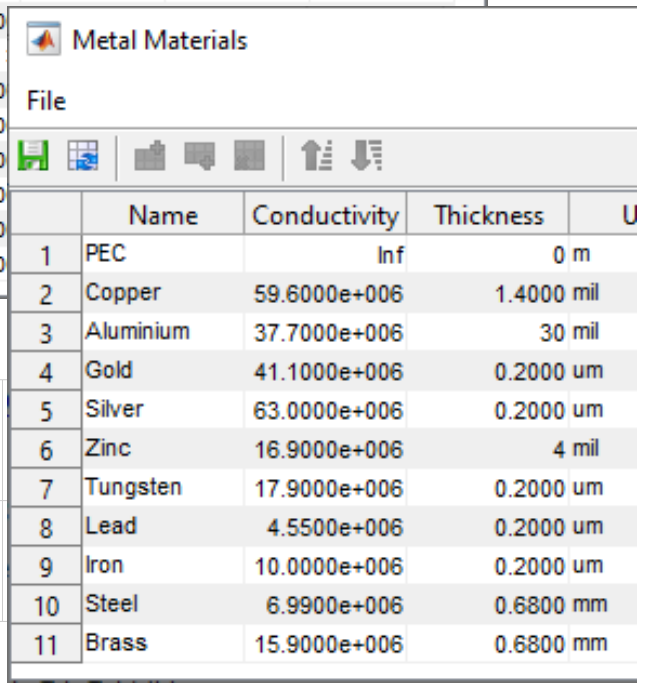
- Use the dielectric catalogue of existing materials
- Define your own: relative permittivity, loss tangent

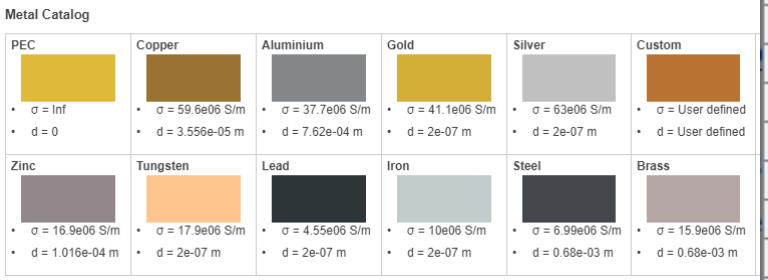
Metal materials:

- Perfect Electrical Conductor (PEC)
  - Zero thickness, infinite conductivity
- Specify the metal material from the catalog
- Define your own: thickness, conductivity

Dielectric	Relative permittivity	Loss Tangent
Air	1	0
Other	>1 (typically <10)	>0 (typically ~1e-3)

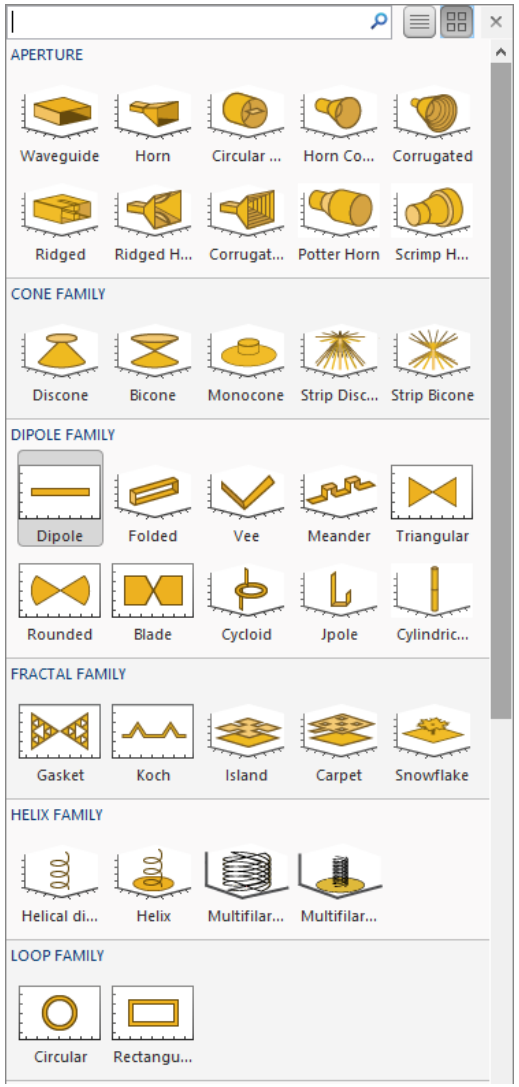


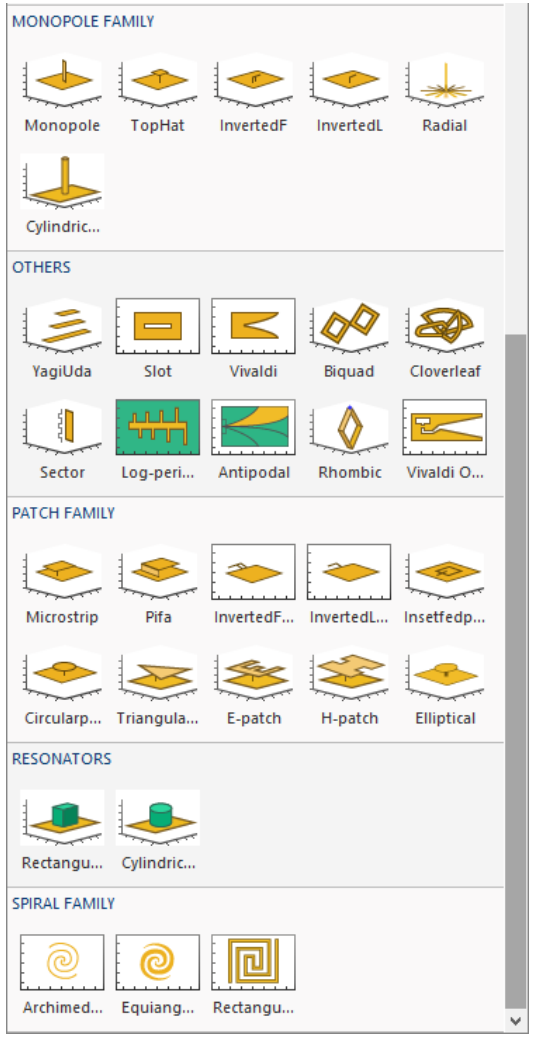




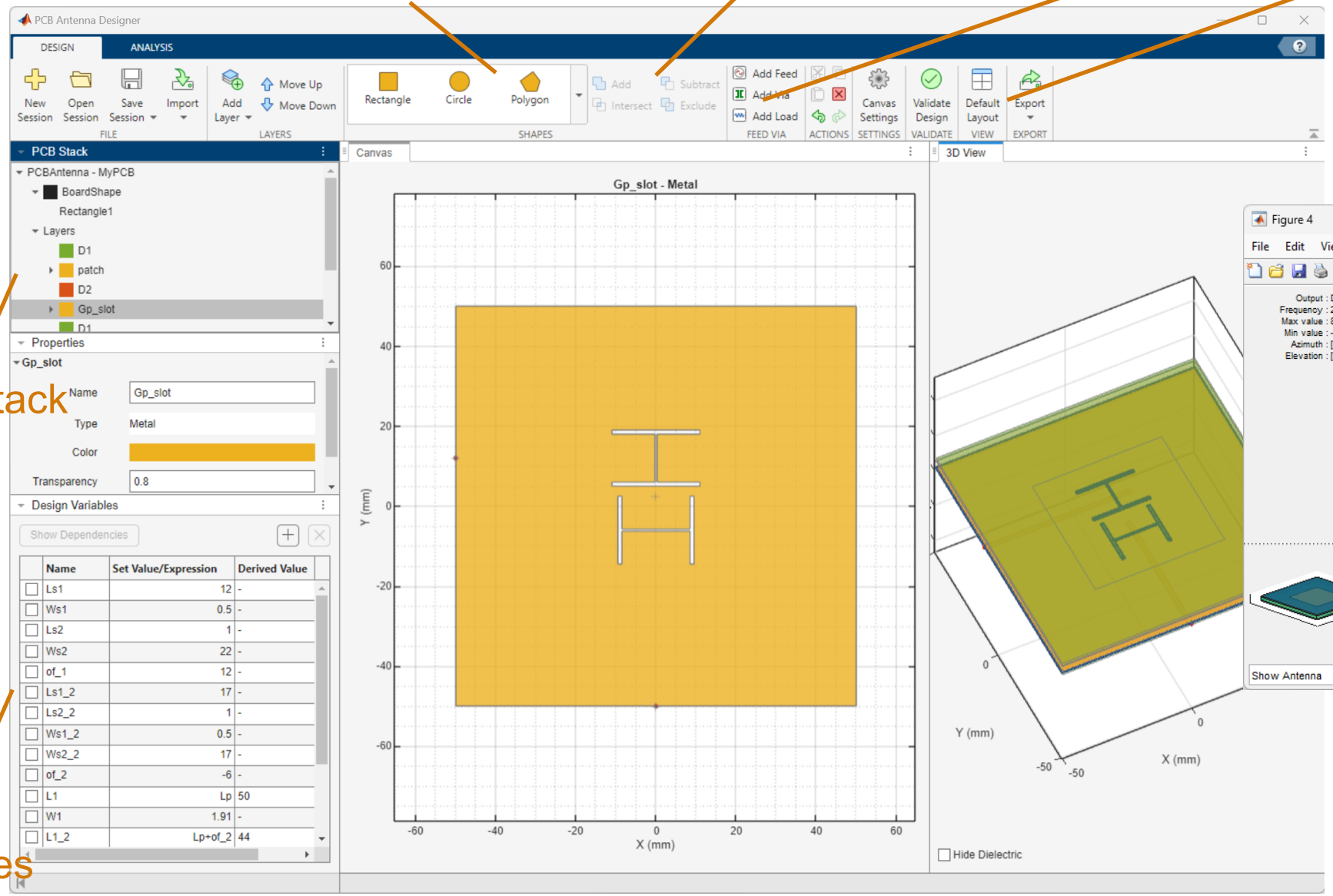
Antenna Catalog: Available Geometries

- Dipole and bowtie antennas
- Monopole antennas
- Patch antennas
- Spiral and loop antennas
- Slot antennas
- Helix antennas
- Fractal antennas
- Waveguide antennas
- Horn and cone antennas
- Other common antennas
- Backing structures

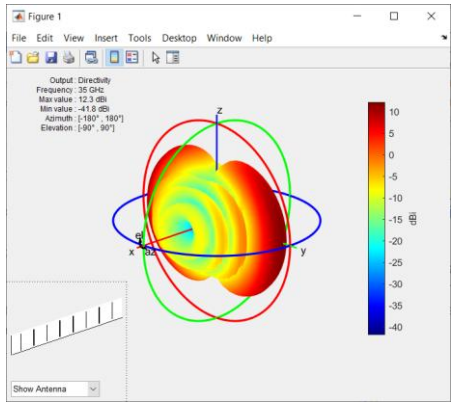


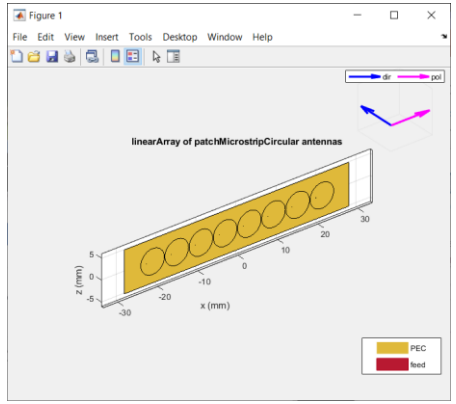


Custom Design of PCB Antennas



Antenna Array Simulation: Integrate Antennas at the System-Level





Instantaneous voltage excitation to compute radiation pattern

V_{TX} I_{TX}

S_{array}

TX Array

- S-parameters
- Embedded element pattern
- Frequency dependent response

TX_{signal} *(EIRP)*

RX_{signal}

Incident plane wave excitation to compute feed currents

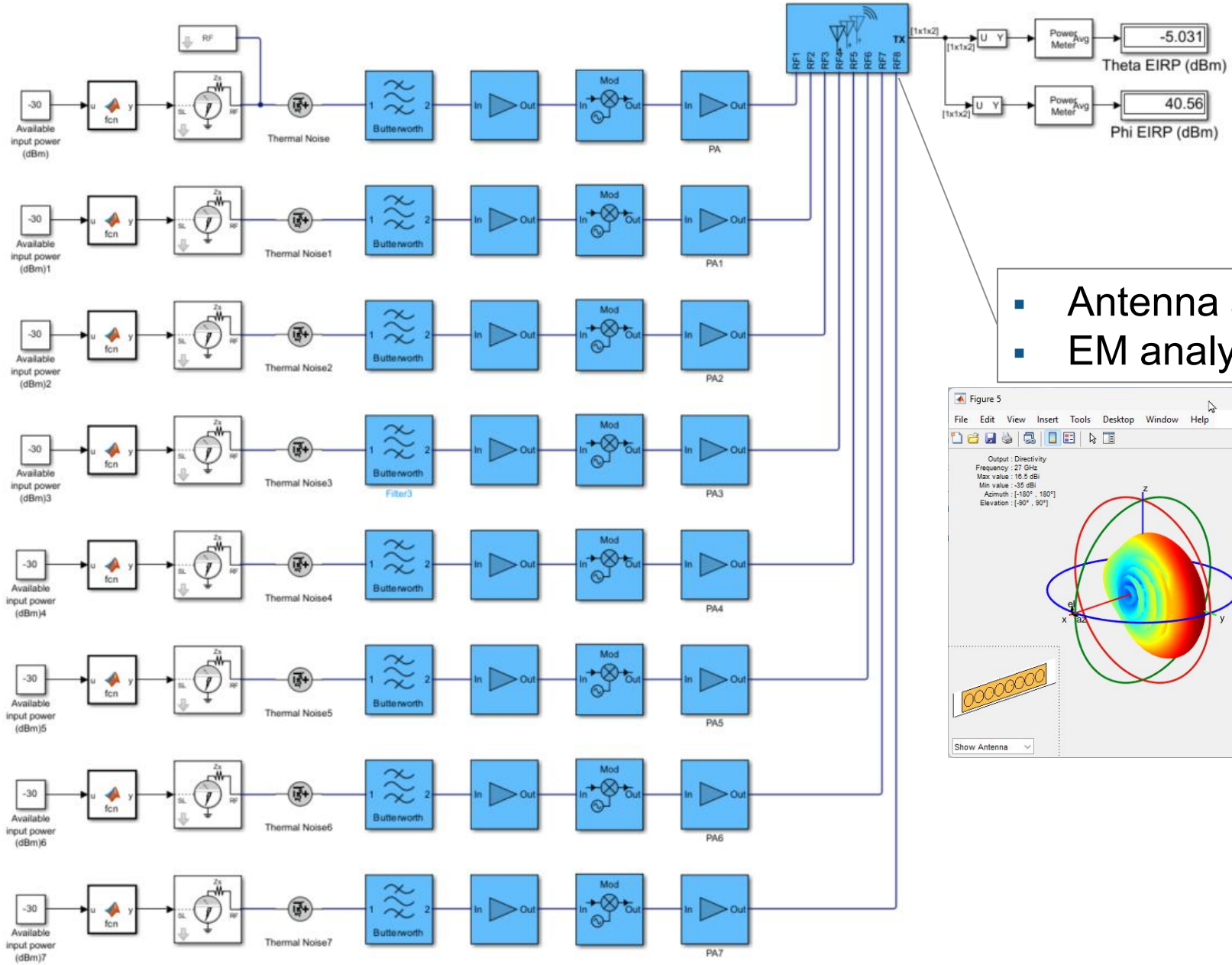
RX Array

- S-parameters
- Embedded element pattern
- Frequency dependent response

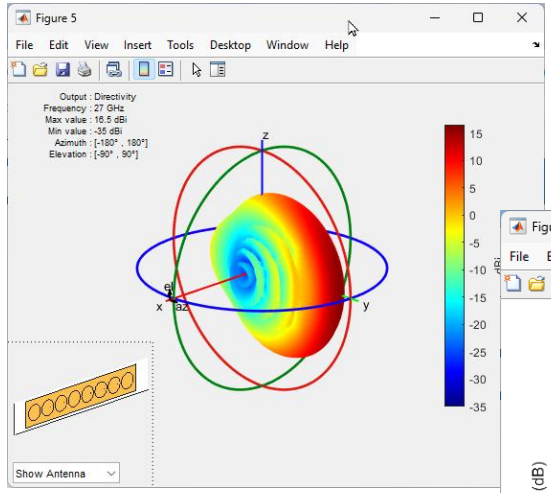
I_{RX} V_{RX}

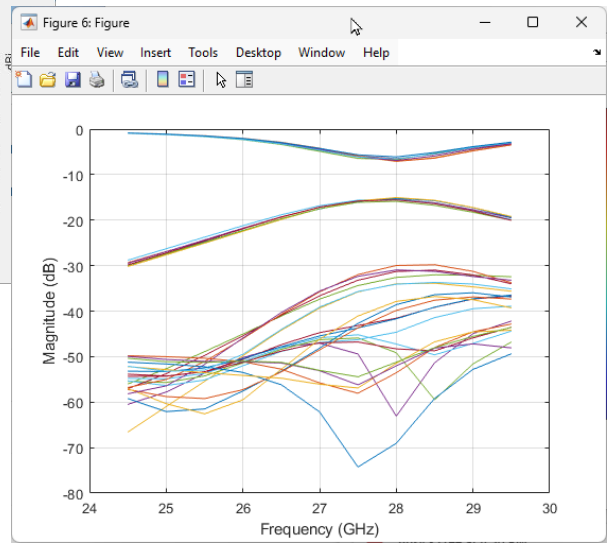
S_{array}

Near and Far Field Coupling Effects on EIRP



- Antenna array S-parameters
- EM analysis of embedded element pattern

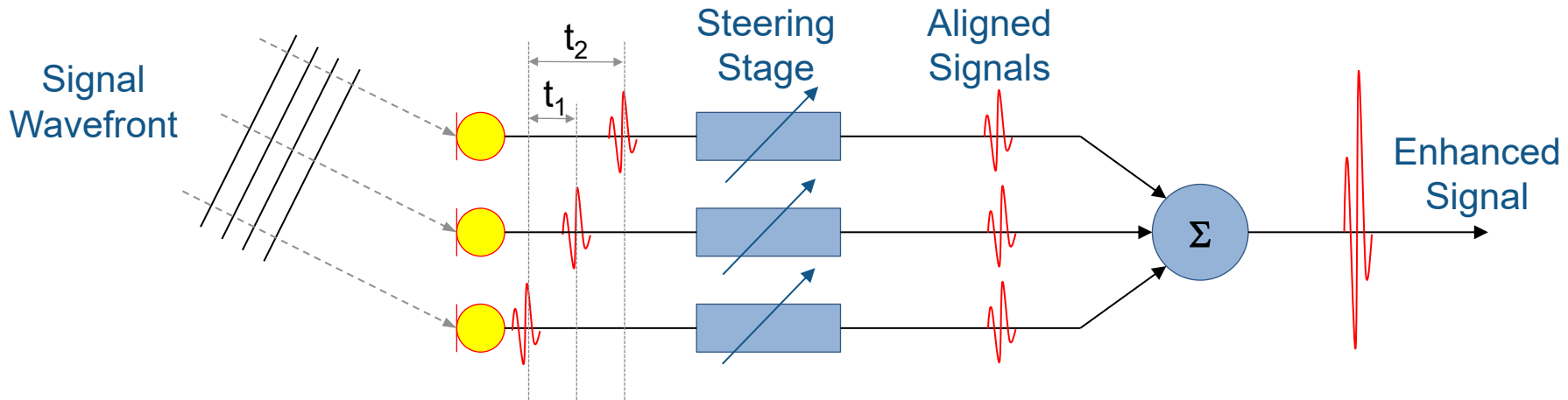




Part 3C — Enabling Beamsteering Algorithms

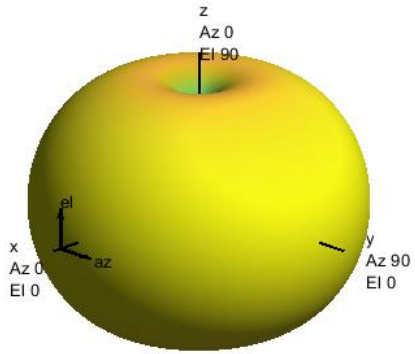
Phased Array Algorithms for Beamsteering and More …

- Beamforming: narrowband and broadband
  - Conventional, MVDR (Capon), LCMV, Frost, time delay, time delay LCMV, subband phase shift, generalized sidelobe canceler
- Direction of arrival estimation
  - Sum and difference monopulse, Beamscan, MVDR (Capon), ESPRIT, Root MUSIC, etc
- Space-time adaptive processing
  - Displaced phase center array (DPCA), Adaptive DPCA (ADPCA), Sample matrix inversion (SMI) , Angle-Doppler response



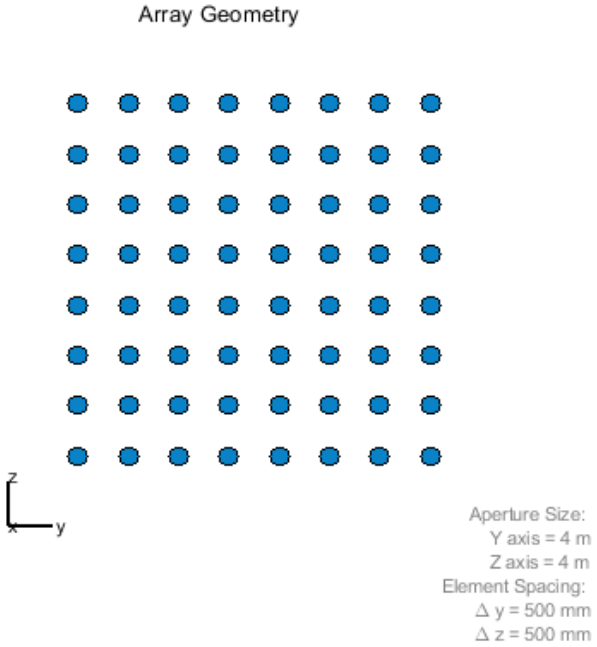
Array Pattern Computed from Pattern Multiplication (Superposition)

Rapidly estimate array gain, beamwidth as a function of array geometry and element spacing

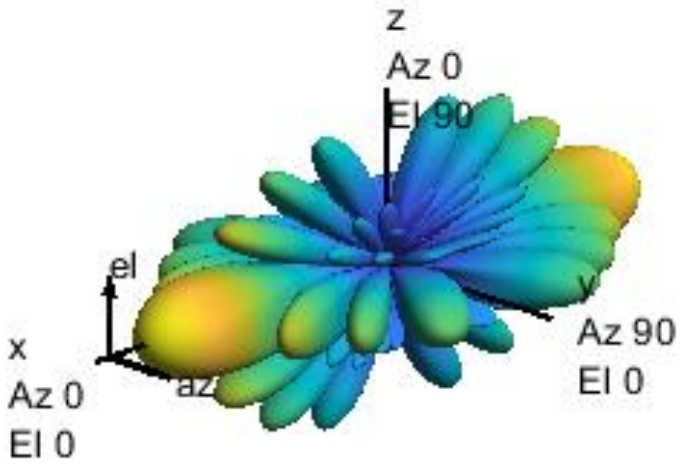


Pattern of (isolated) antenna element

×

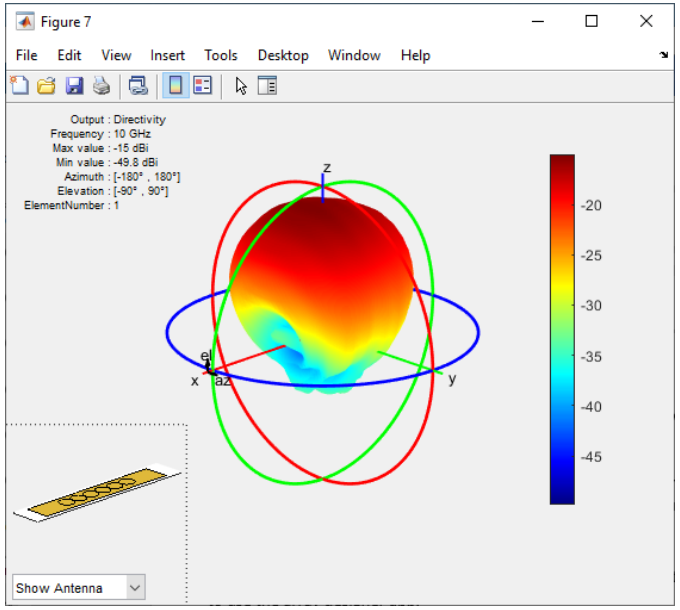


=



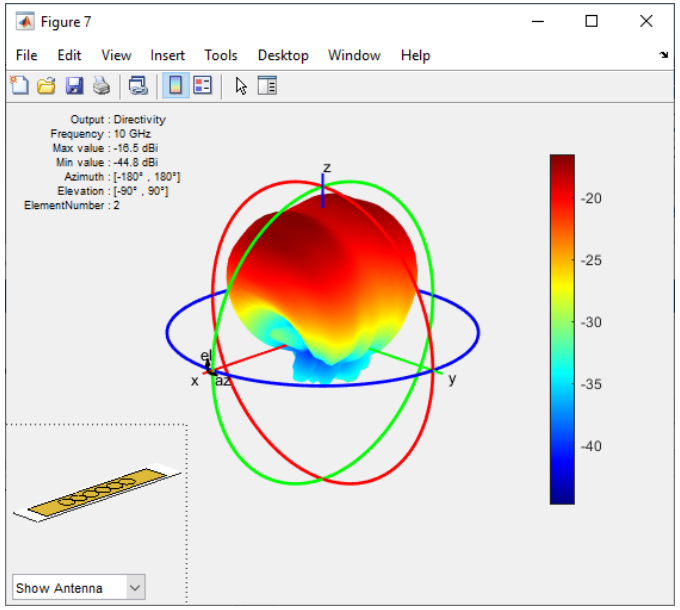
Coupling and Edge Effects Impact the Antenna Far-Field Pattern

Embedded element #1



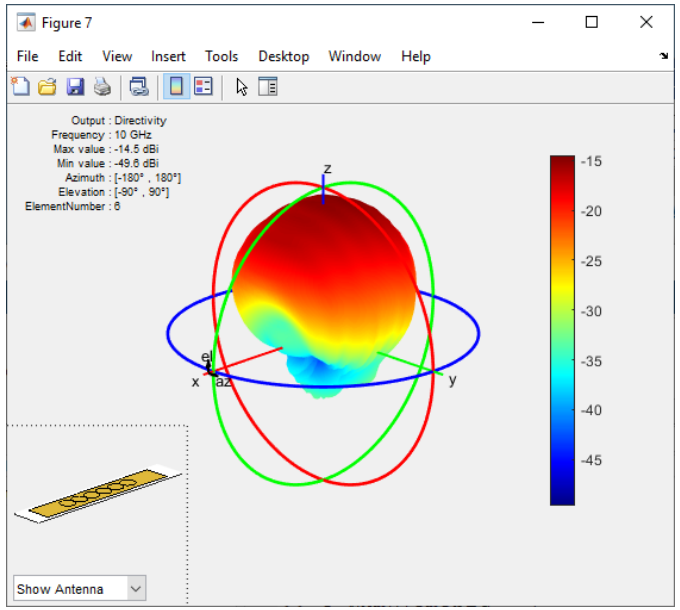

```
pattern(1,10e9,...
'ElementNumber',1);
```

Embedded element #2




```
pattern(1,10e9,...
'ElementNumber',2);
```

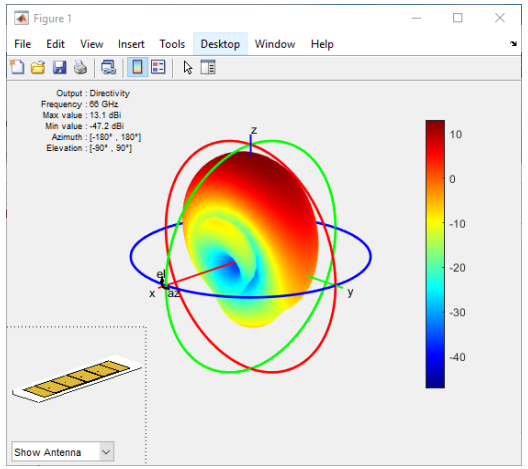
Embedded element #6



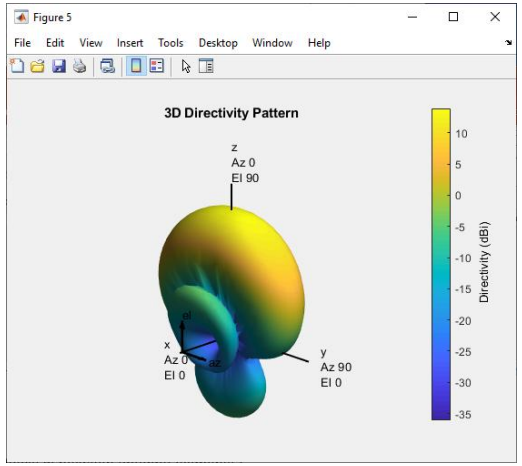

```
pattern(1,10e9,...
'ElementNumber',6);
```

Pattern Superposition of Isolated Element Might Not Be Accurate

Full wave analysis

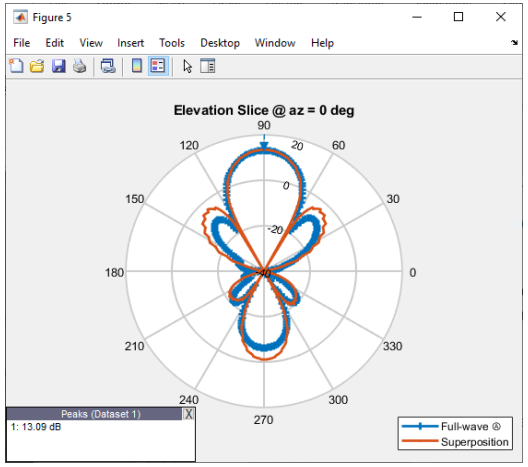


Isolated element pattern superposition

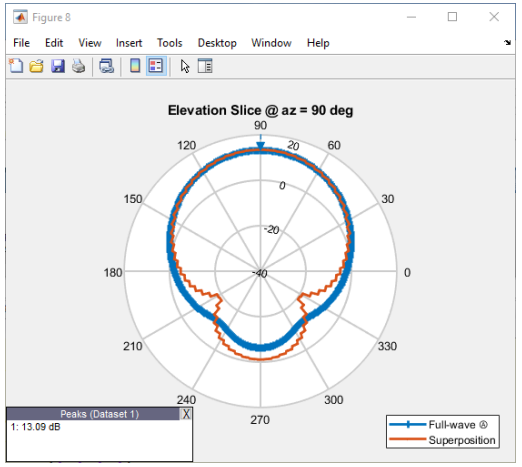


Comparison

Azimuth = 0 deg



Azimuth = 90 deg

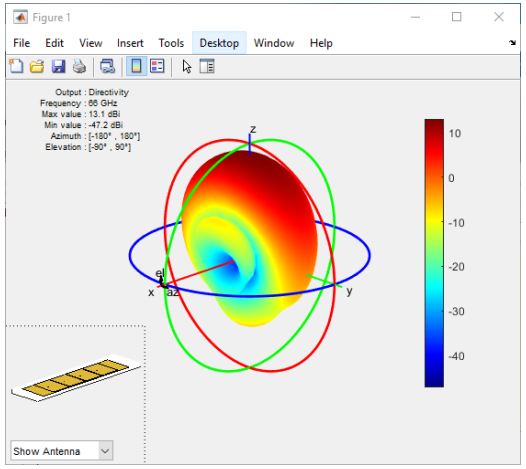


Blue = Full wave

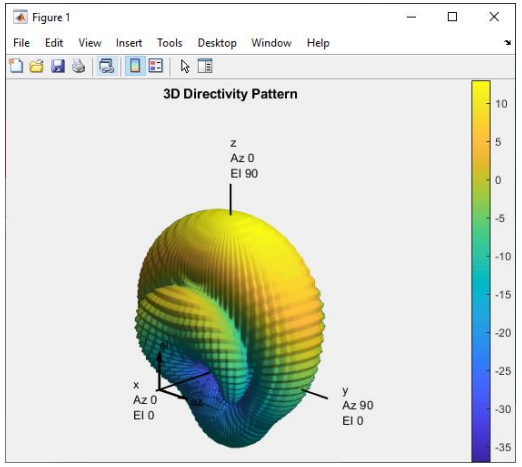
Red = Pattern superposition

Using the Embedded Element Pattern Provides More Accurate Results

Full wave analysis

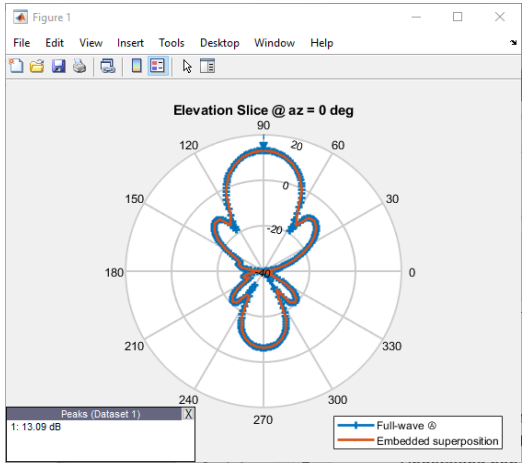


Embedded element pattern superposition

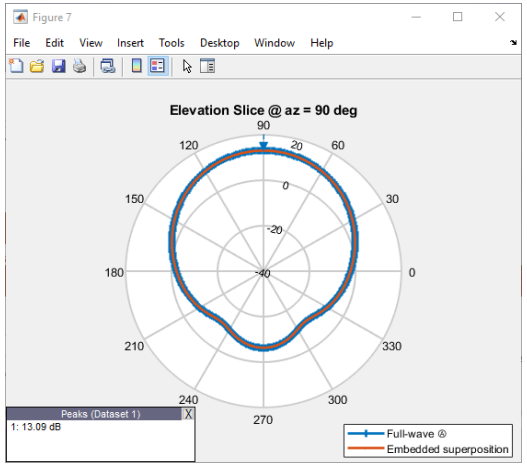


Comparison

Azimuth = 0 deg



Azimuth = 90 deg



Blue = Full wave

Red = Pattern superposition

Compute Phase Shifts for Analog Beamformer Simulation

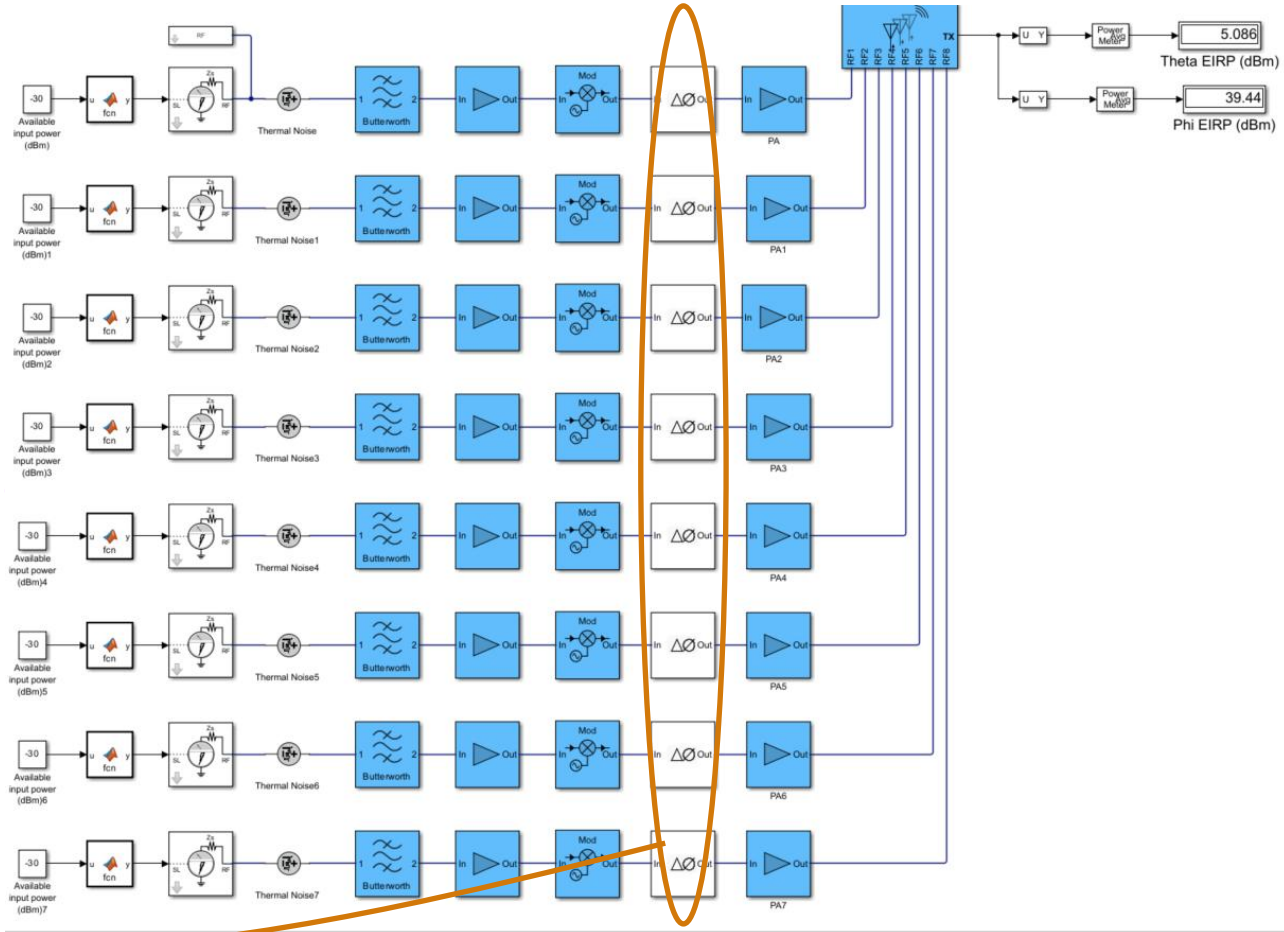
beamformingPhaseShifts.m

```
patchAnt = phased.CustomAntennaElement;
patchAnt.AzimuthAngles = (-180:5:180);
patchAnt.ElevationAngles = (-90:5:90);
patchAnt.MagnitudePattern = p;
patchAnt.PhasePattern = zeros(37,73);
patchAnt.MatchArrayNormal = false;

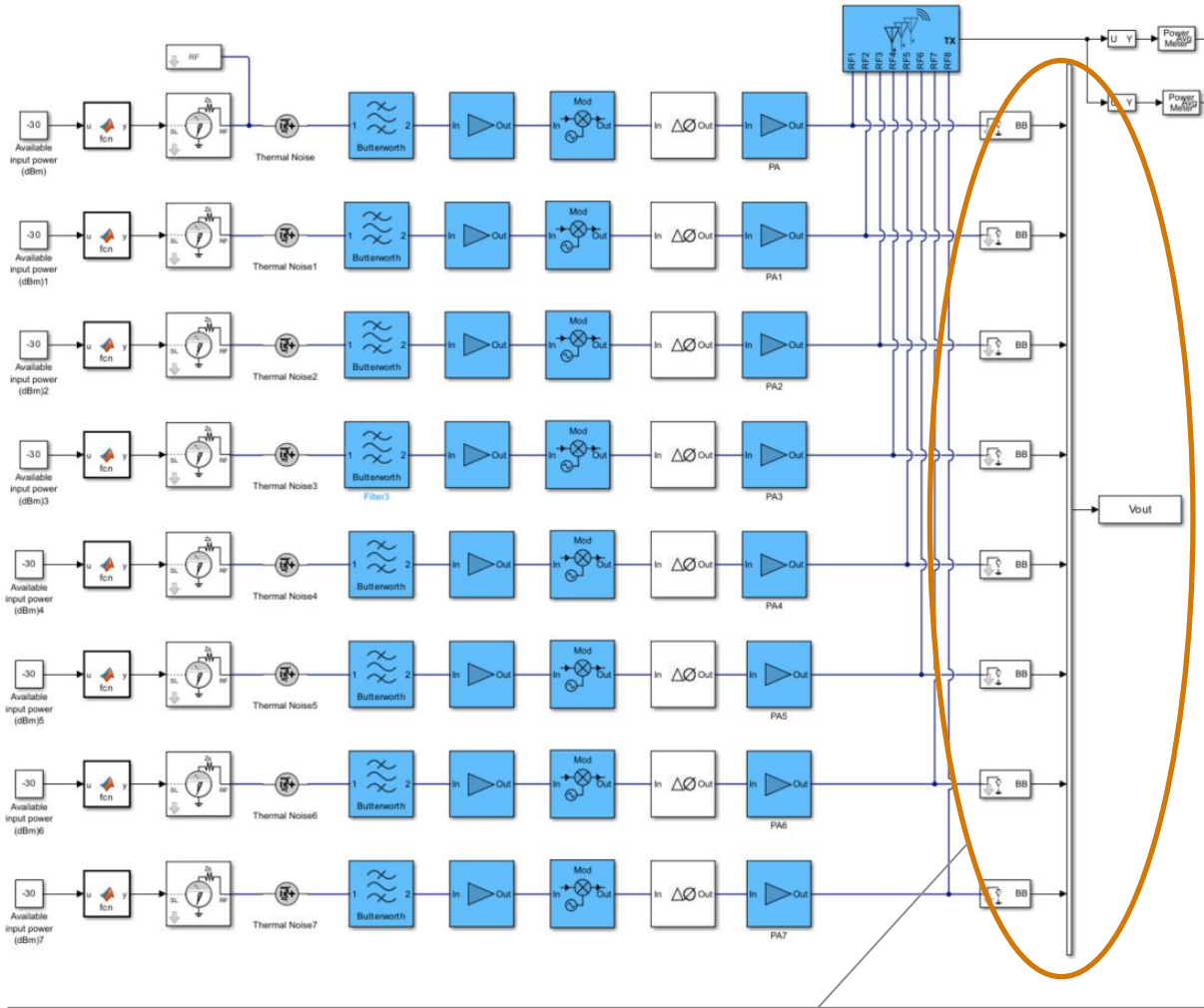
array = phased.ULA('NumElements',8, ...
    'ElementSpacing', 0.0064,...
    'ArrayAxis','x');
array.Element = patchAnt;
Direction = 90+30;

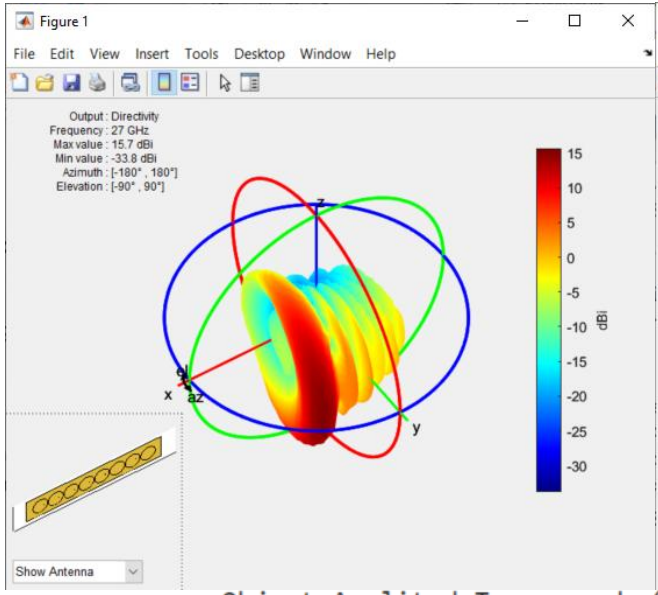
beamformer = phased.PhaseShiftBeamformer(..
    'SensorArray',array,...
    'Direction',[Direction; 0],...
    'OperatingFrequency',27e9, ...
    'WeightsOutputPort',true);
[~, phaseShifts] = beamformer(ones(1, 8));

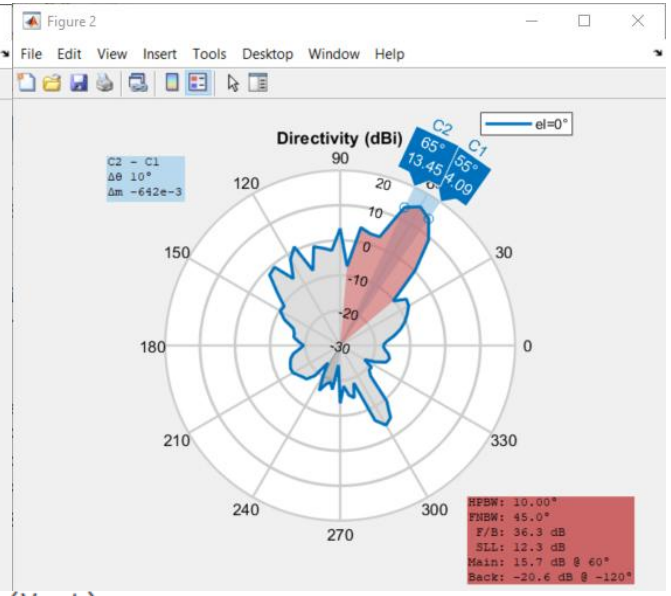
% Normalize weights between 0 and 360
phaseShifts = angle(phaseShifts)/pi*180;
```



Plot Array Pattern as a Function of Actual Excitation







- Save (complex) voltage excitation at antenna port
- Apply amplitude tapers and phase shifts
- Compute and plot pattern

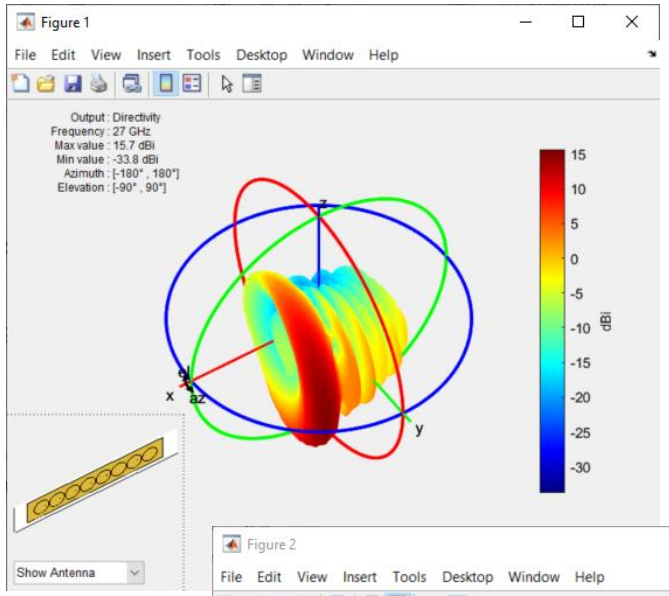
```
arrayObject.AmplitudeTaper = abs(Vout);
arrayObject.PhaseShift = angle(Vout)/pi*180;
```

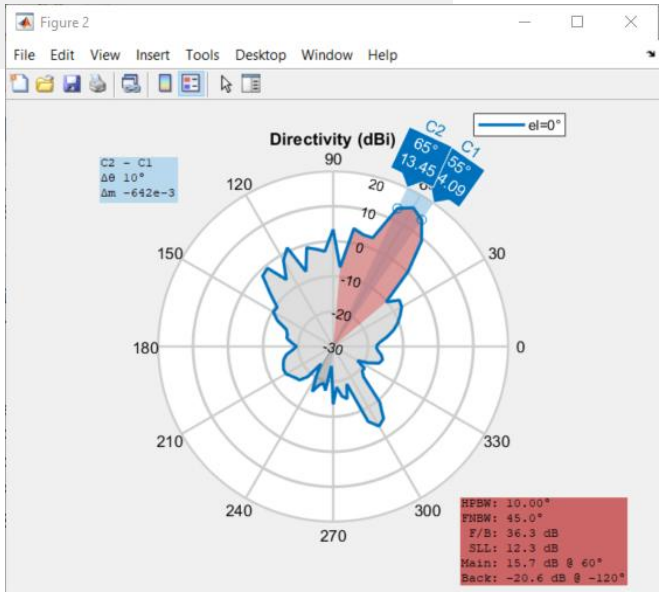
```
%% Array Analysis
plotFrequency = 27e9;
figure;
pattern(arrayObject, plotFrequency);
azRange = 0:5:360;
figure;
pattern(arrayObject, plotFrequency, azRange, 0);
```

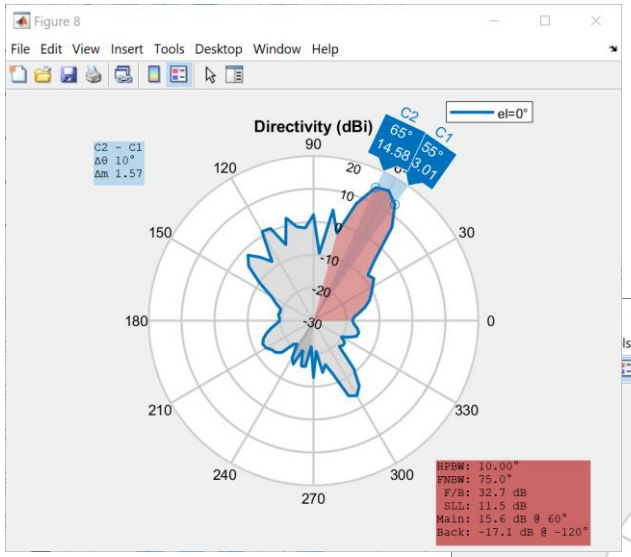
Array Pattern As a Function of PA Output Impedance

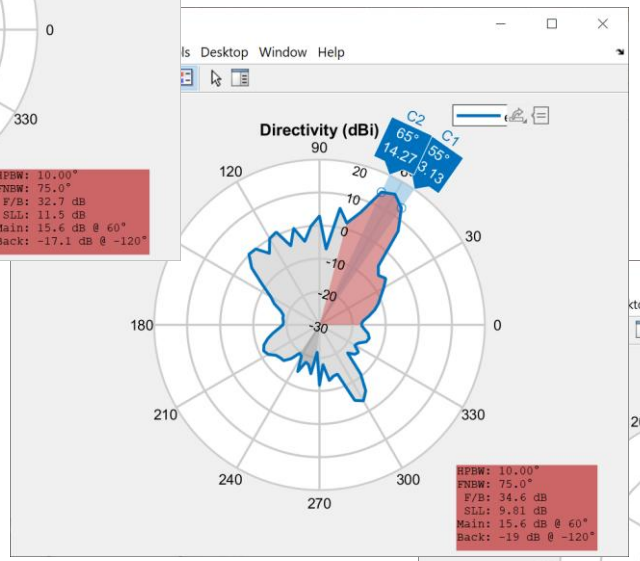
P_{in} = -30dBm

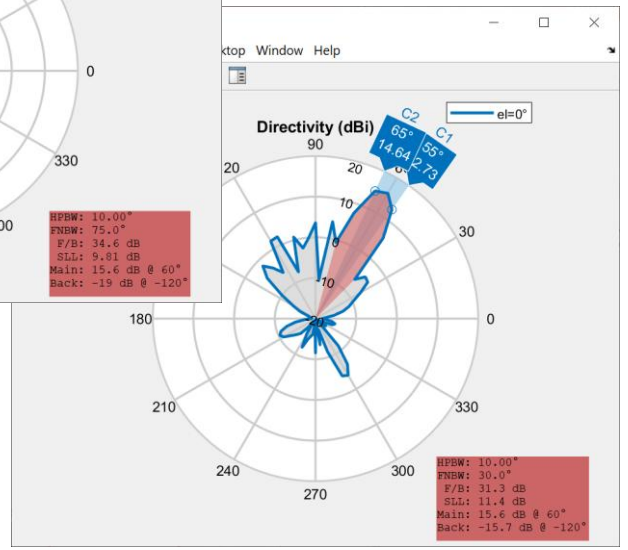
Randomized PA Output Impedance [25..75]Ohm







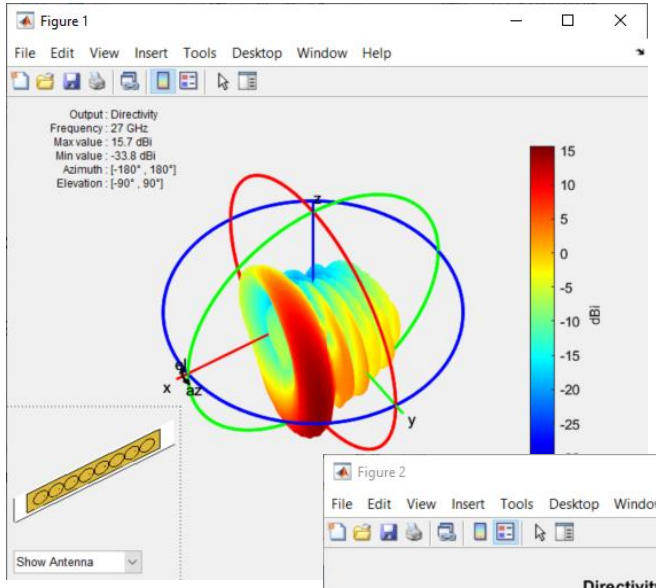


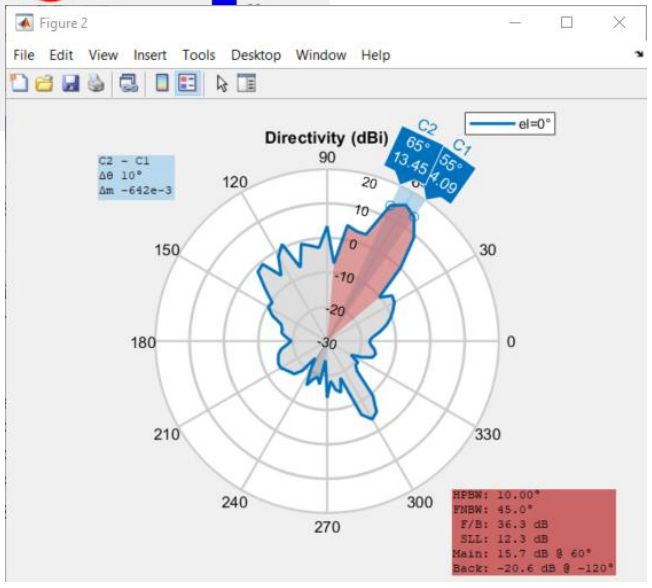


Array Pattern As a Function of Frequency (Beam Squinting)

Reference:

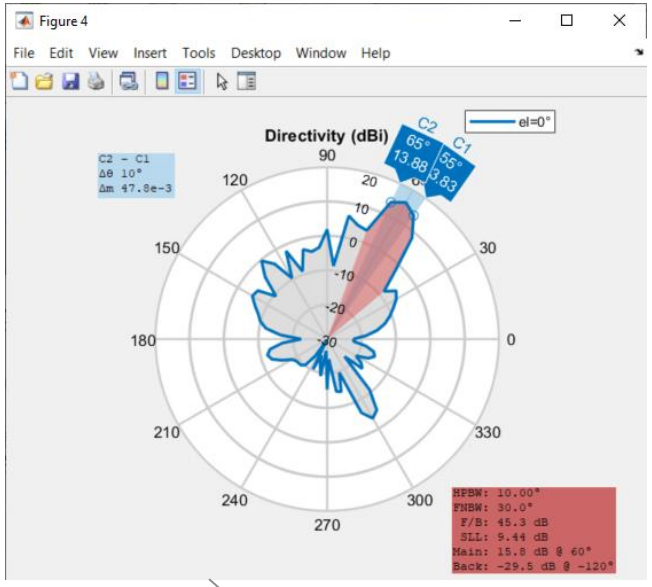
- F_{in} = 4GHz
- F_{out} = 27GHz





Input frequency offset:

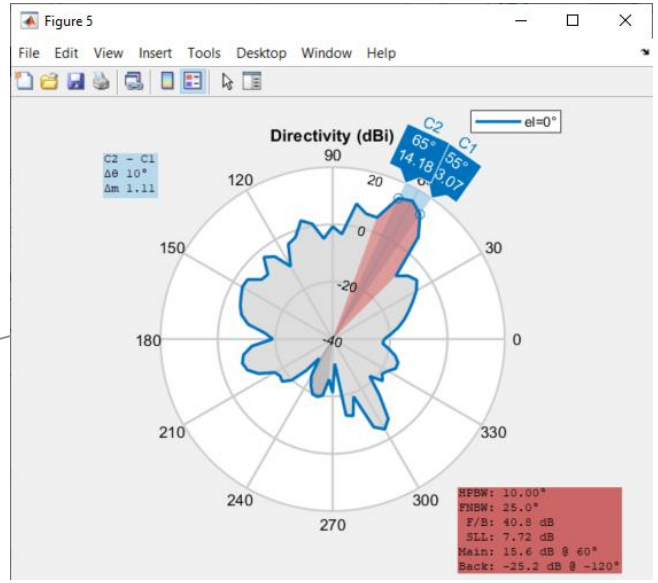
- F_{in} = 4.5GHz
- F_{out} = 27.5GHz



- Pattern distortion due to frequency dispersion

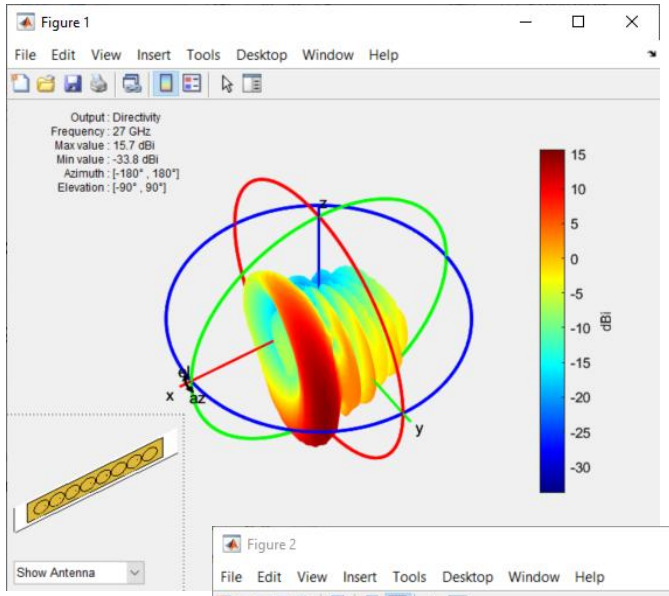
LO frequency offset:

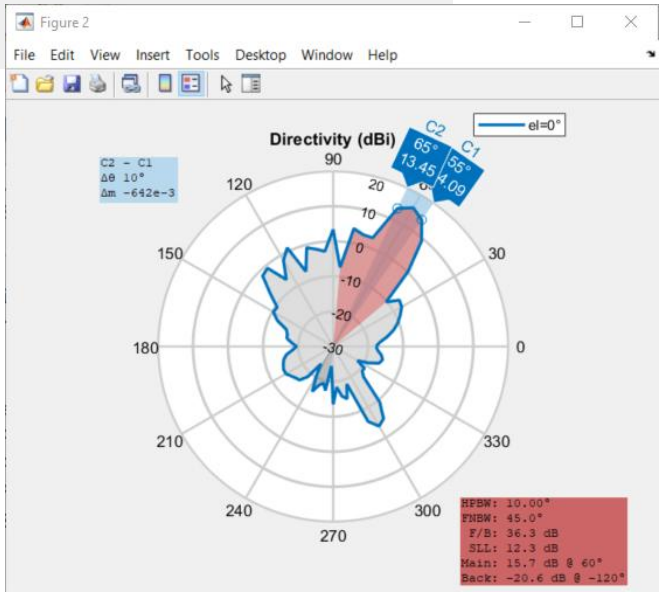
- F_{in} = 4GHz
- F_{out} = 28GHz



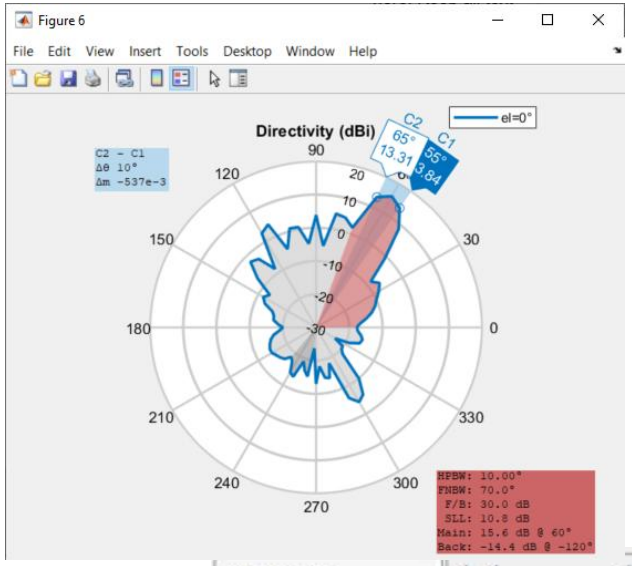
Array Pattern As a Function of Input Power and Non-Linearity

P_{in} = -30dBm





Saturation: P_{in} = 0dBm

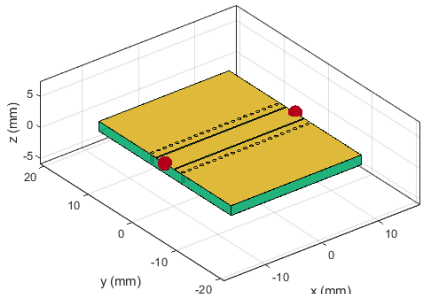


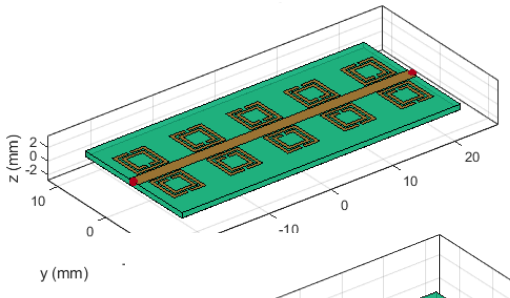
- Pattern distortion
- Reduced EIRP

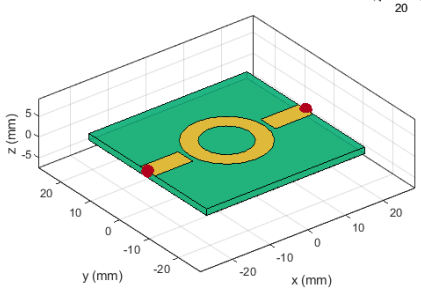
Cascade	1..1	1..2	1..3	1..4	1..5	EIRP:54.9467
Fout (GHz)	4	4	27	27	27	Directivity:9.738
Friis-Pout (dBm)	0	14.8273	20.8273	45.8273	45.2089	
HB-Pout (dBm)	0	14.3986	19.7763	37.0390	36.4205	
Friis-GainT (dB)	0	14.8273	20.8273	45.8273	45.2089	
HB-GainT (dB)	0	14.3986	19.7763	37.0390	36.4205	
Friis-NF (dB)	0	2.9000	2.9724	3.0260	3.0260	
HB-NF (dB)	0	2.6879	2.2233	0.3045	0.3045	
Friis-OIP3 (dBm)	Inf	28	29.8756	44.5746	43.9562	
HB-OIP3 (dBm)	Inf	25.9700	27.5221	38.2007	37.5823	

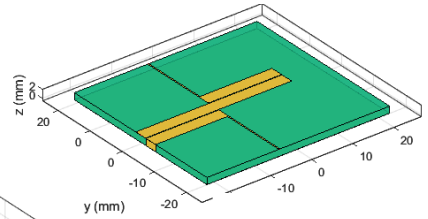
Part 3D — Power Divider PCB Design

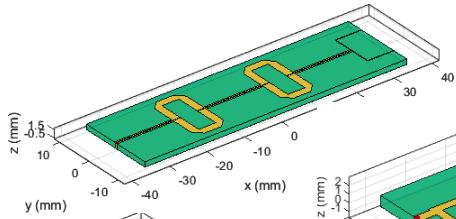
RF PCB Components

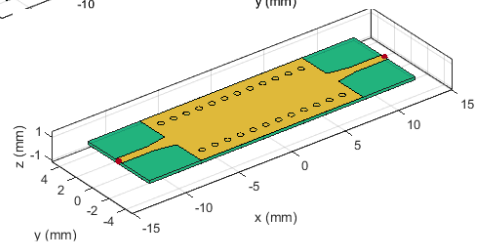


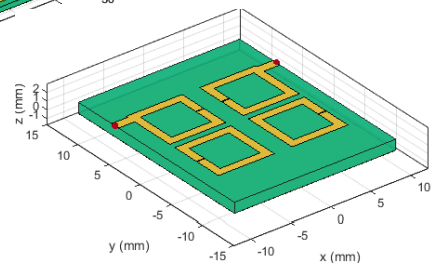


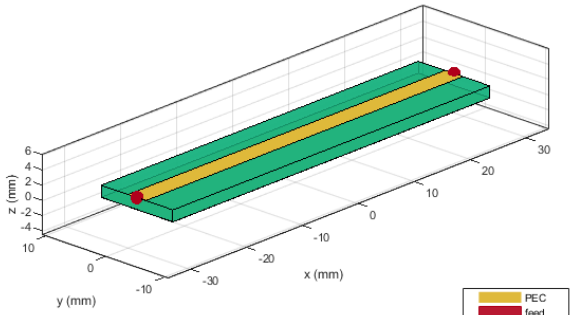


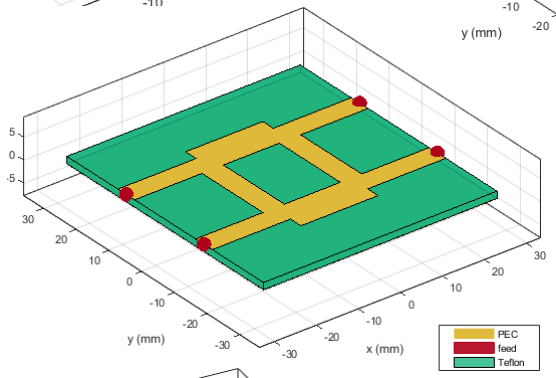


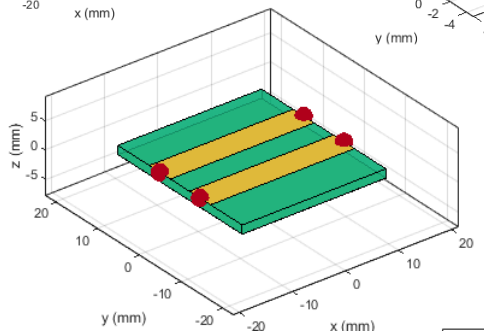


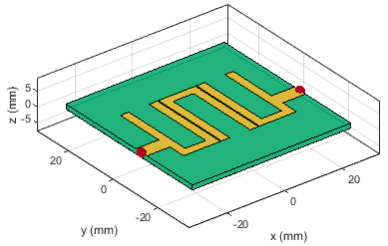


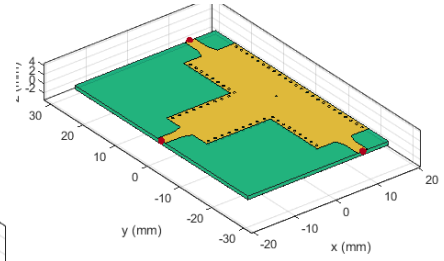


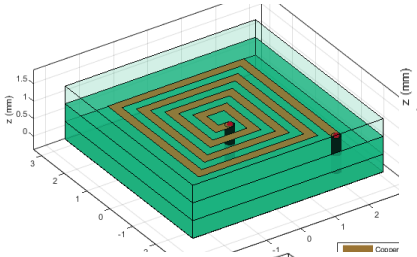


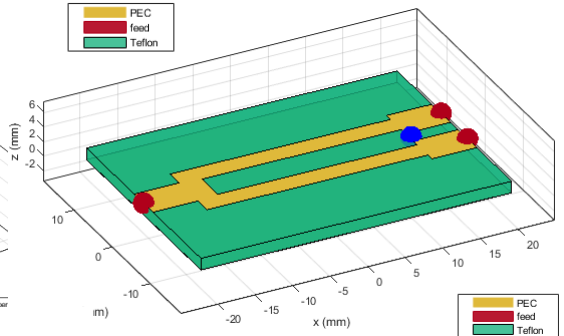


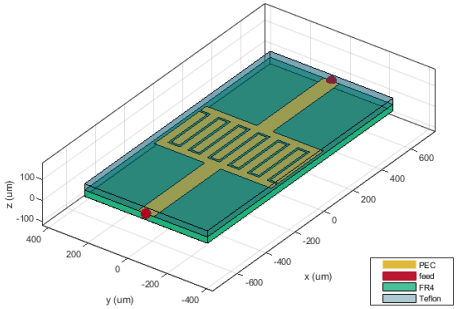


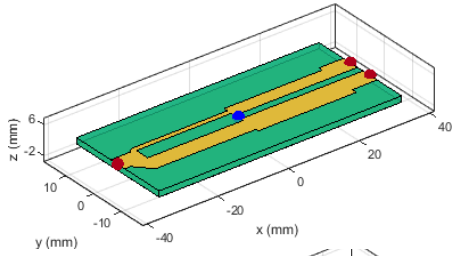


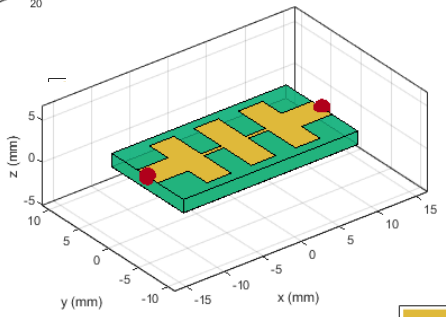


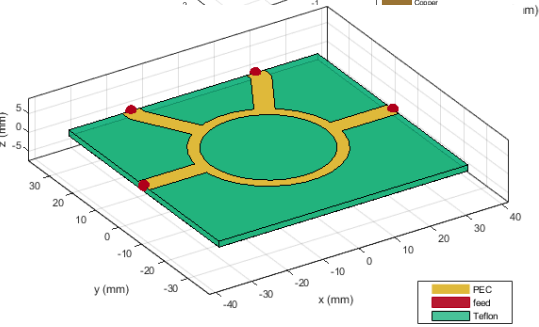


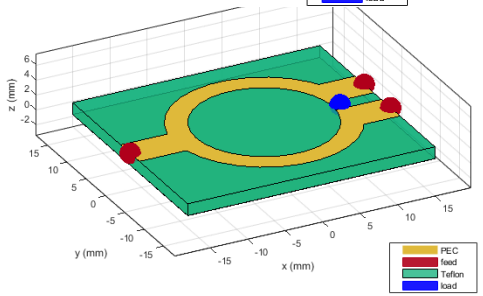


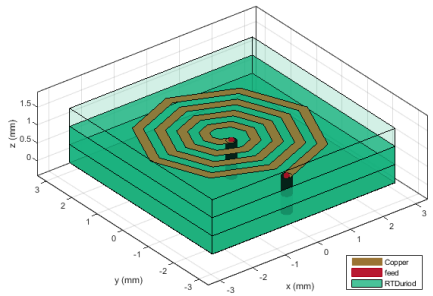


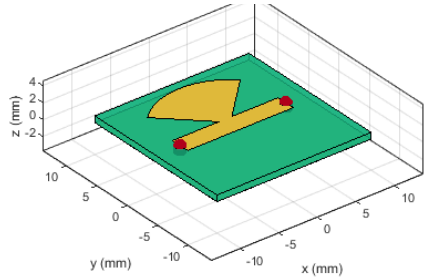


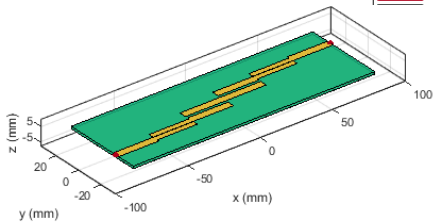


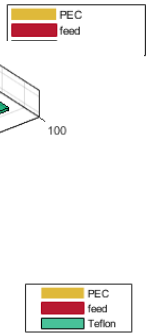






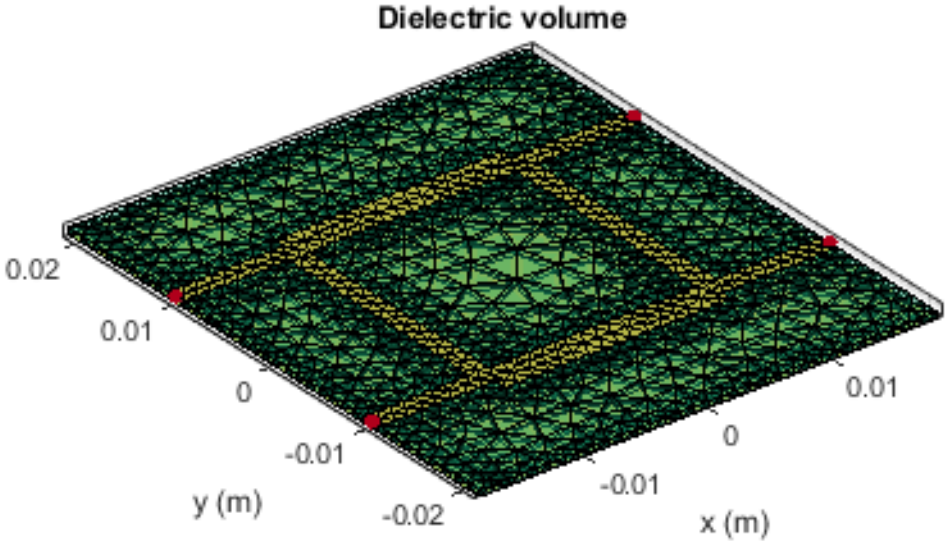






Electromagnetic Analysis of PCB Structures

- Full-wave solver: Method of Moments
- Volumetric mesh of the PCB
  - Analysis-driven automatic mesh generation
  - Manual control on meshing
- Specify material properties: metal and dielectric



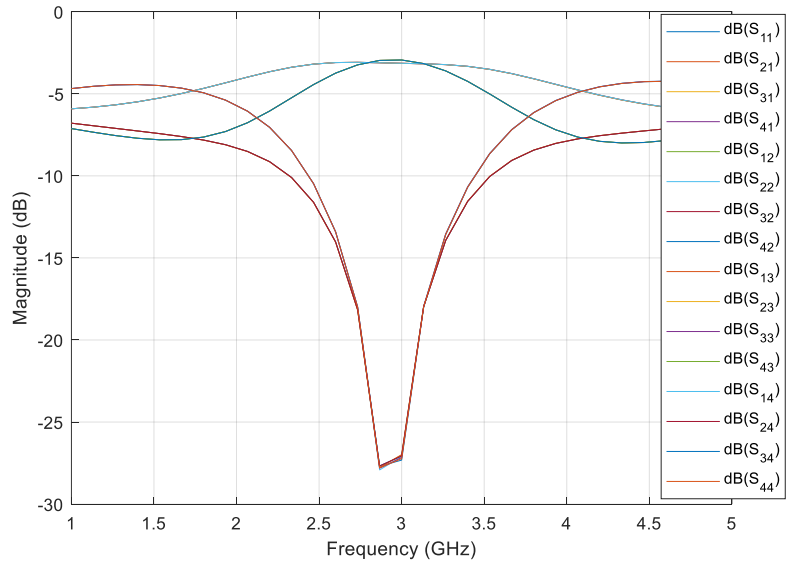
Dielectric Materials

File

	Name	Relative_Permittivity	Loss_Tangent	Frequency
1	Air	1	0	1.0000e+00
2	FR4	4.8000	0.0260	100.0000e+0.
3	Teflon	2.1000	2.0000e-04	100.0000e+0.
4	Foam	1.0300	1.5000e-04	50.0000e+00
5	Polystyrene	2.5500	1.0000e-04	100.0000e+0.
6	Mica	3	1.0000e-04	100.0000e+0.
7	Plexiglas	2.5900	0.0068	10.0000e+00
8	Fused quartz	3.7800	1.0000e-04	10.0000e+009
9	E glass	6.2200	0.0023	100.0000e+0...
10	RO4725JXR	2.5500	0.0022	2.5000e+009
11	RO4730JXR	3	0.0023	2.5000e+009
12	TMM3	3.4500	0.0020	10.0000e+009
13	TMM4	4.7000	0.0020	10.0000e+009
14	TMM6	6.3000	0.0023	10.0000e+009
15	TMM10	9.8000	0.0022	10.0000e+009
16	TMM10i	9.9000	0.0020	10.0000e+009
17	Taconic RF-35	3.5000	0.0018	1.9000e+009

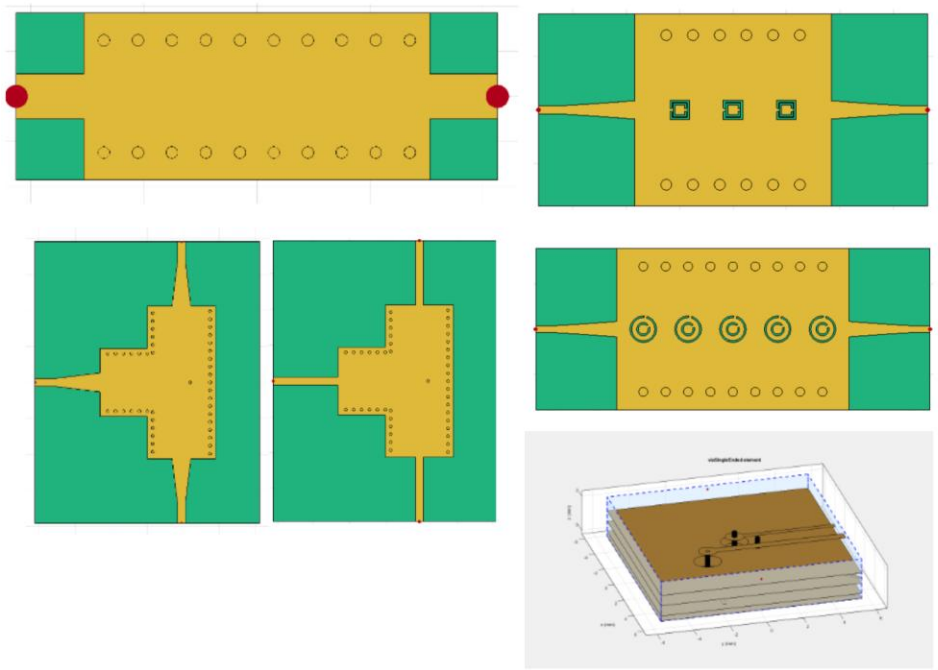
Metal Catalog

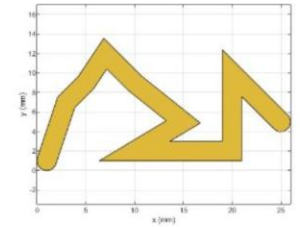
PEC	Copper	Aluminium	Gold	Silver	Custom
σ = Inf d = 0	σ = 59.6e06 S/m d = 3.556e-05 m	σ = 37.7e06 S/m d = 7.62e-04 m	σ = 41.1e06 S/m d = 2e-07 m	σ = 63e06 S/m d = 2e-07 m	σ = User defined d = User defined
Zinc	**Tungsten**	**Lead**	**Iron**	**Steel**	**Brass**
σ = 16.9e06 S/m d = 1.016e-04 m	σ = 17.9e06 S/m d = 2e-07 m	σ = 4.55e06 S/m d = 2e-07 m	σ = 10e06 S/m d = 2e-07 m	σ = 6.99e06 S/m d = 0.68e-03 m	σ = 15.9e06 S/m d = 0.68e-03 m

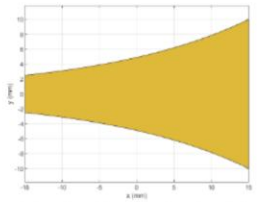


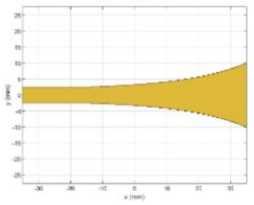
Electromagnetic Modeling of RF PCBs

- Design RF PCBs or import Gerber files
- Extensive catalog of distributed components
- Perform electromagnetic analysis using finite element method
- Calculate S-parameters of PCBs with metal shield

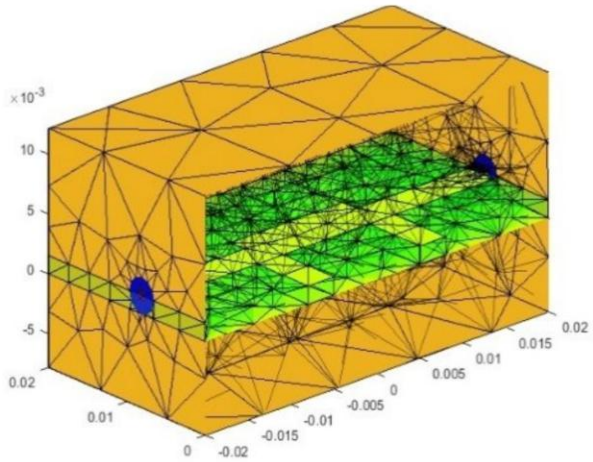




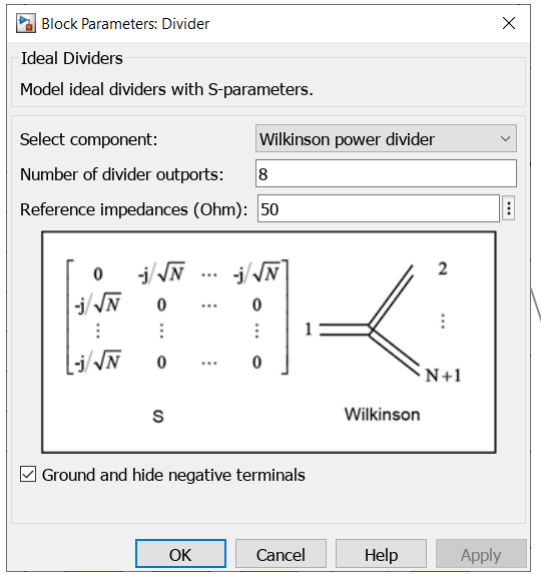


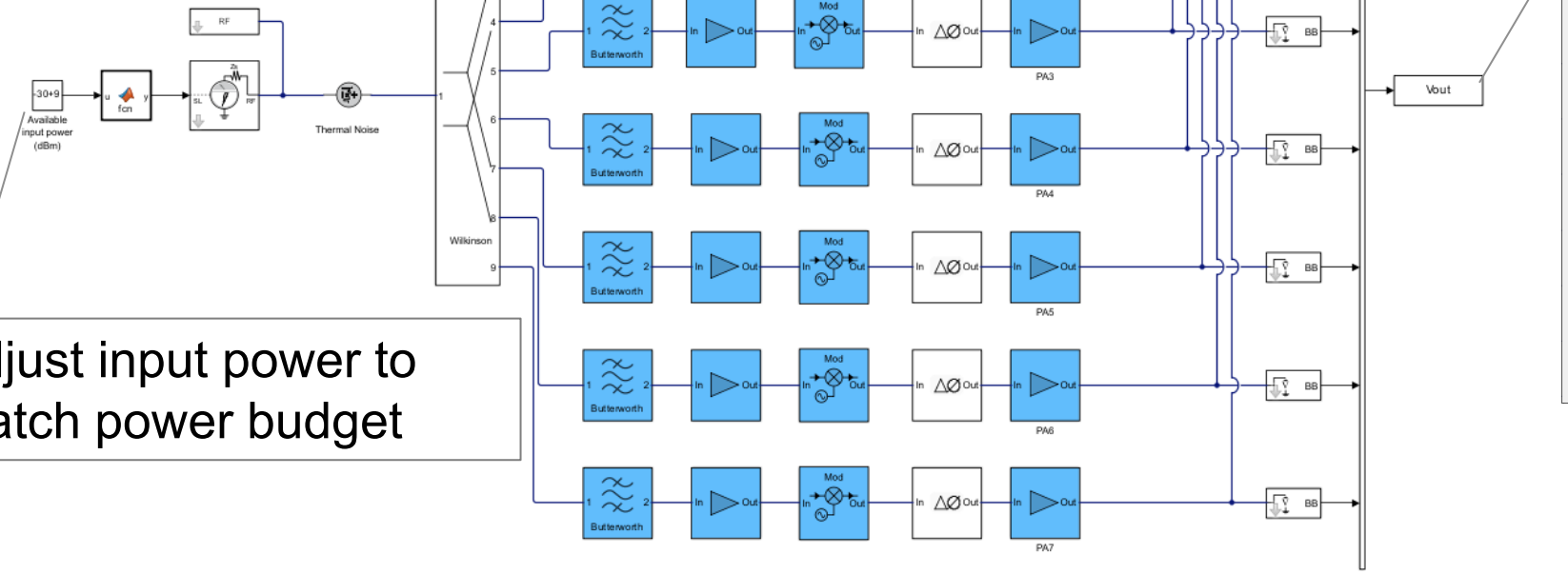


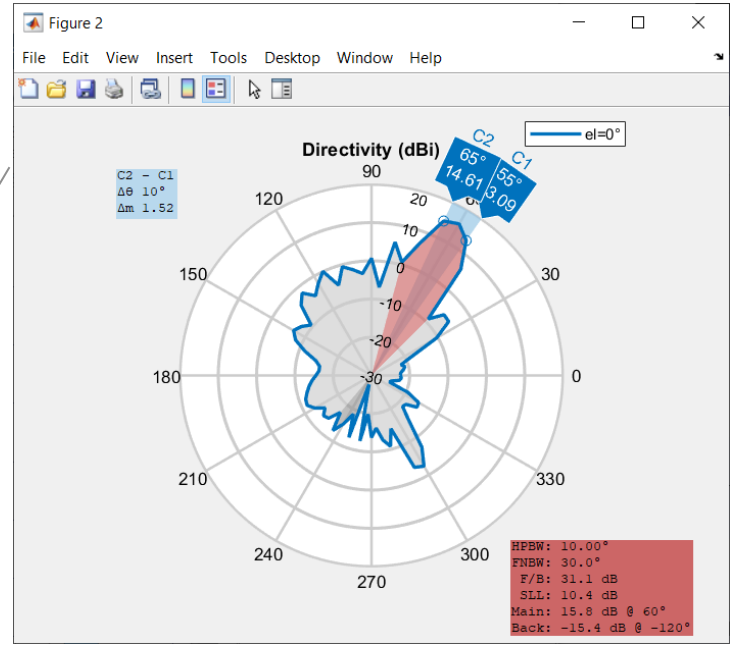
```
branchline = design(couplerBranchLine,3e9);
branchline.IsShielded = true;
```



Ideal Wilkinson Power Splitter



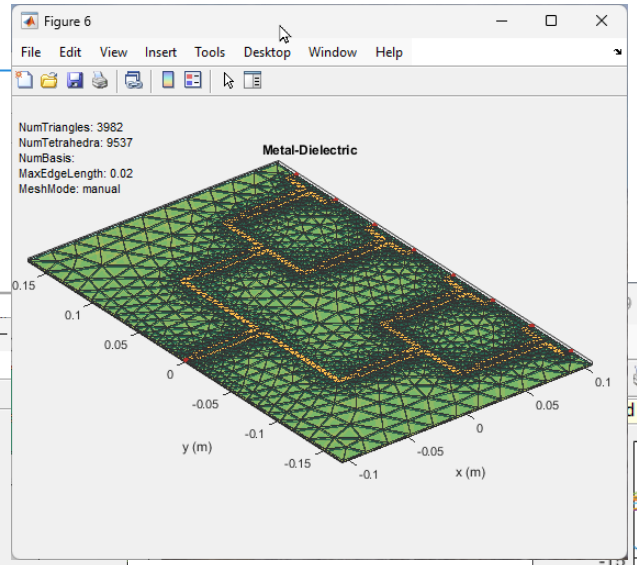


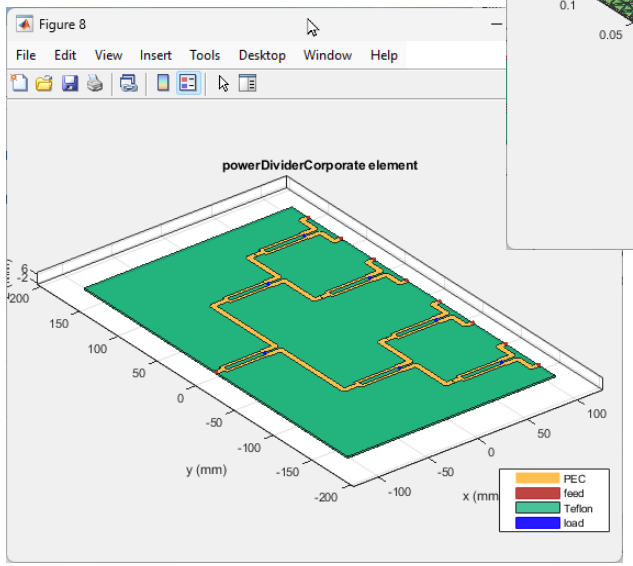


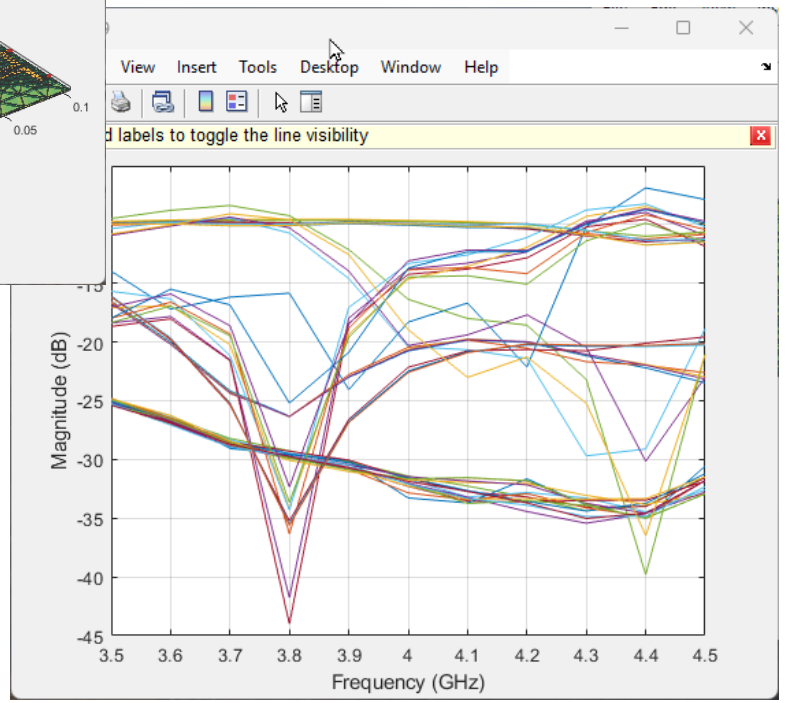
- Adjust input power to match power budget

Design and EM Analysis of PCB Wilkinson Splitter

```
%% Design corporate power divider and perform EM analysis
splitter = design(powerDividerCorporate(NumOutputPorts=8), 4e9);
figure; show(splitter);
figure; mesh(splitter,'MaxEdgeLength',0.02)
%%
f = linspace(3.5e9, 4.5e9, 11);
sSplitter = sparameters(splitter,f);
figure; rfplot(sSplitter)
```





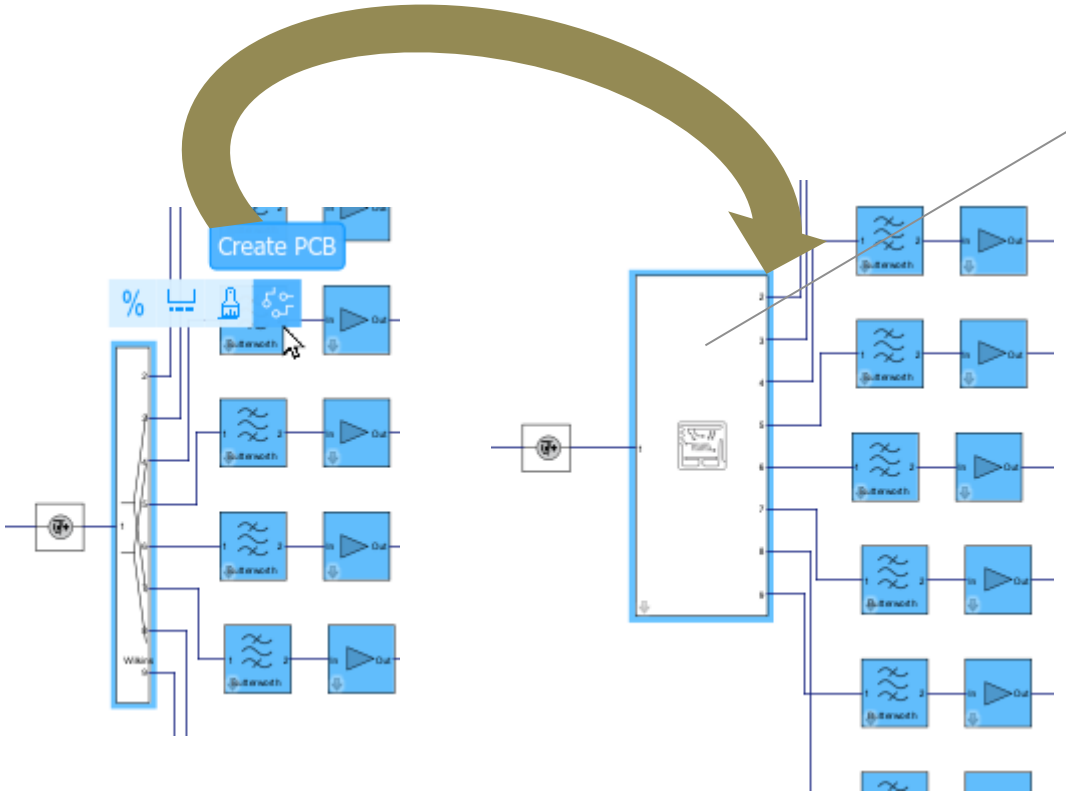


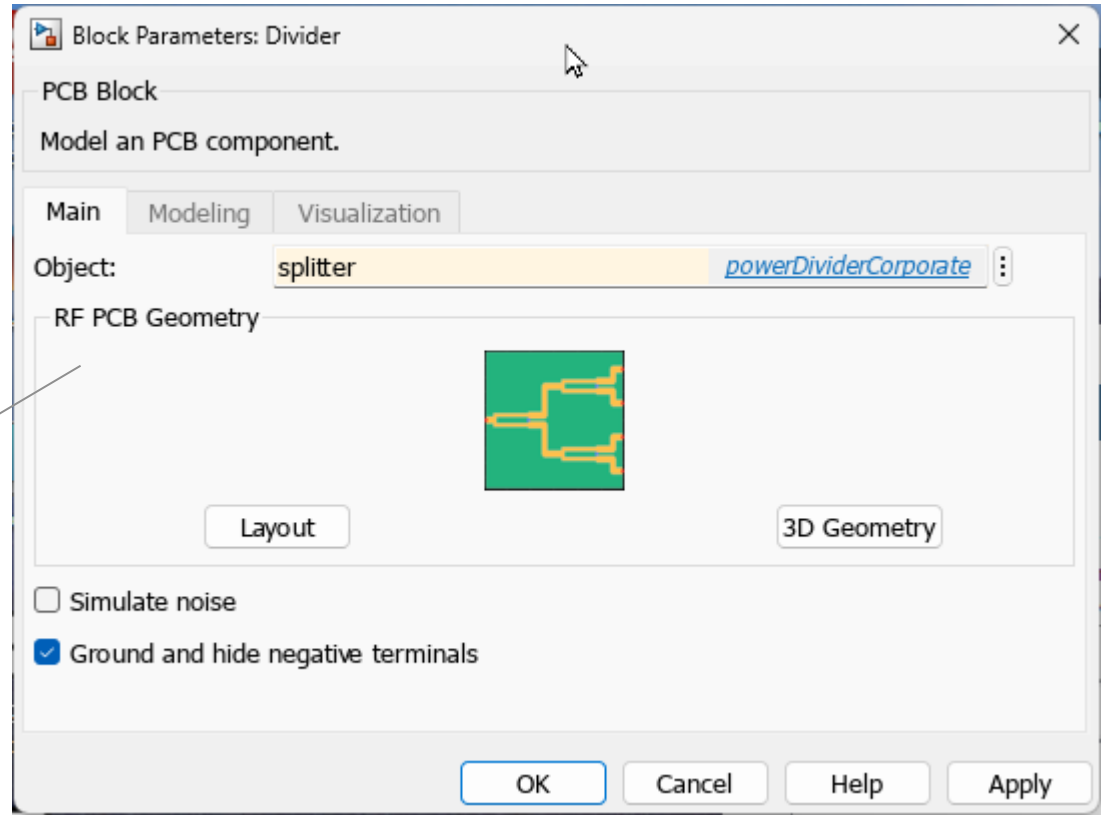
Specify :

- Geometry
- PCB stack
- Material properties

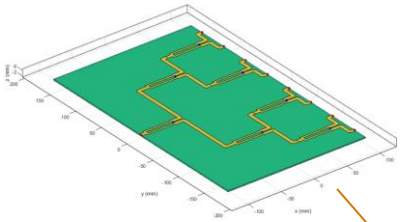
Adding the Wilkinson Splitter in the Array Simulation

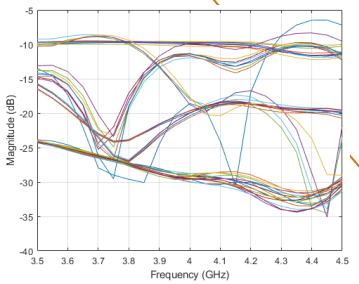
Assisted conversion of PCB components



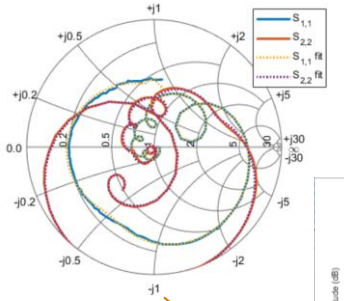


Sources of Dispersion and Impedance Mismatches

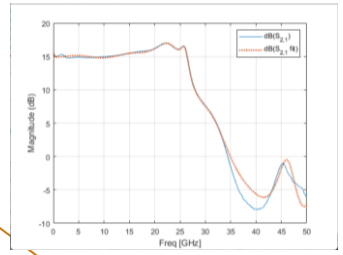


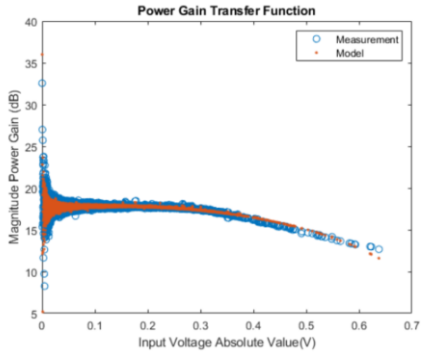


Passive distributed elements

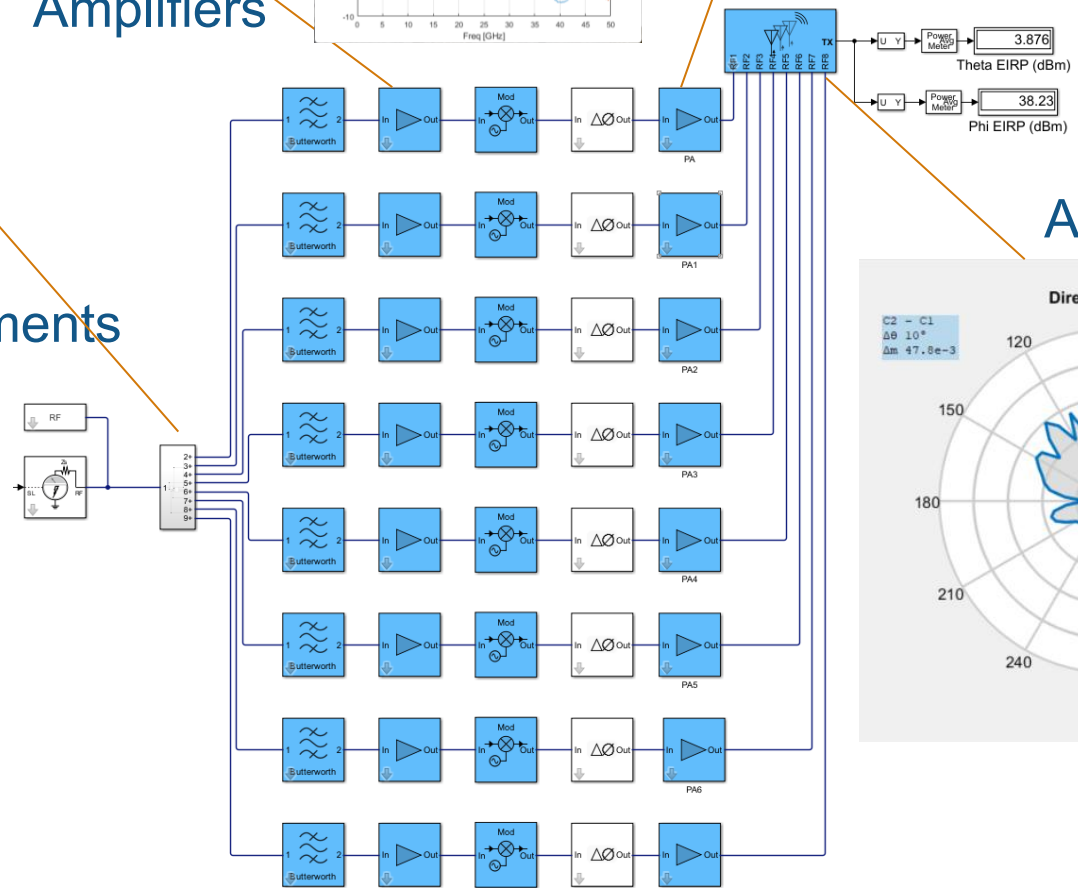


Amplifiers

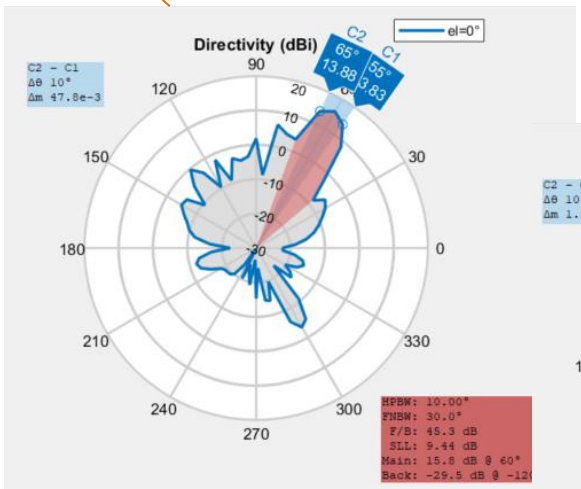


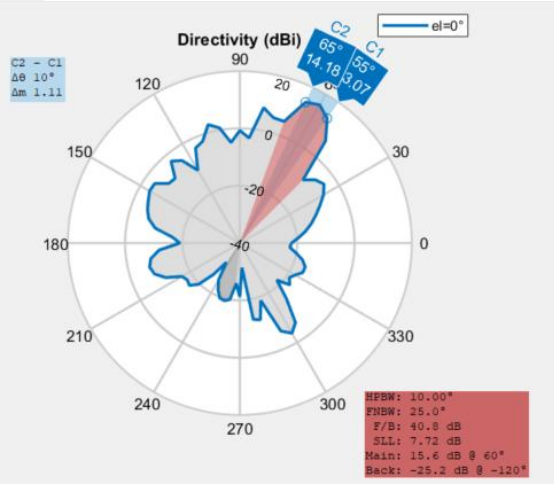


Power amplifiers memory effects



Antennas impedance and pattern

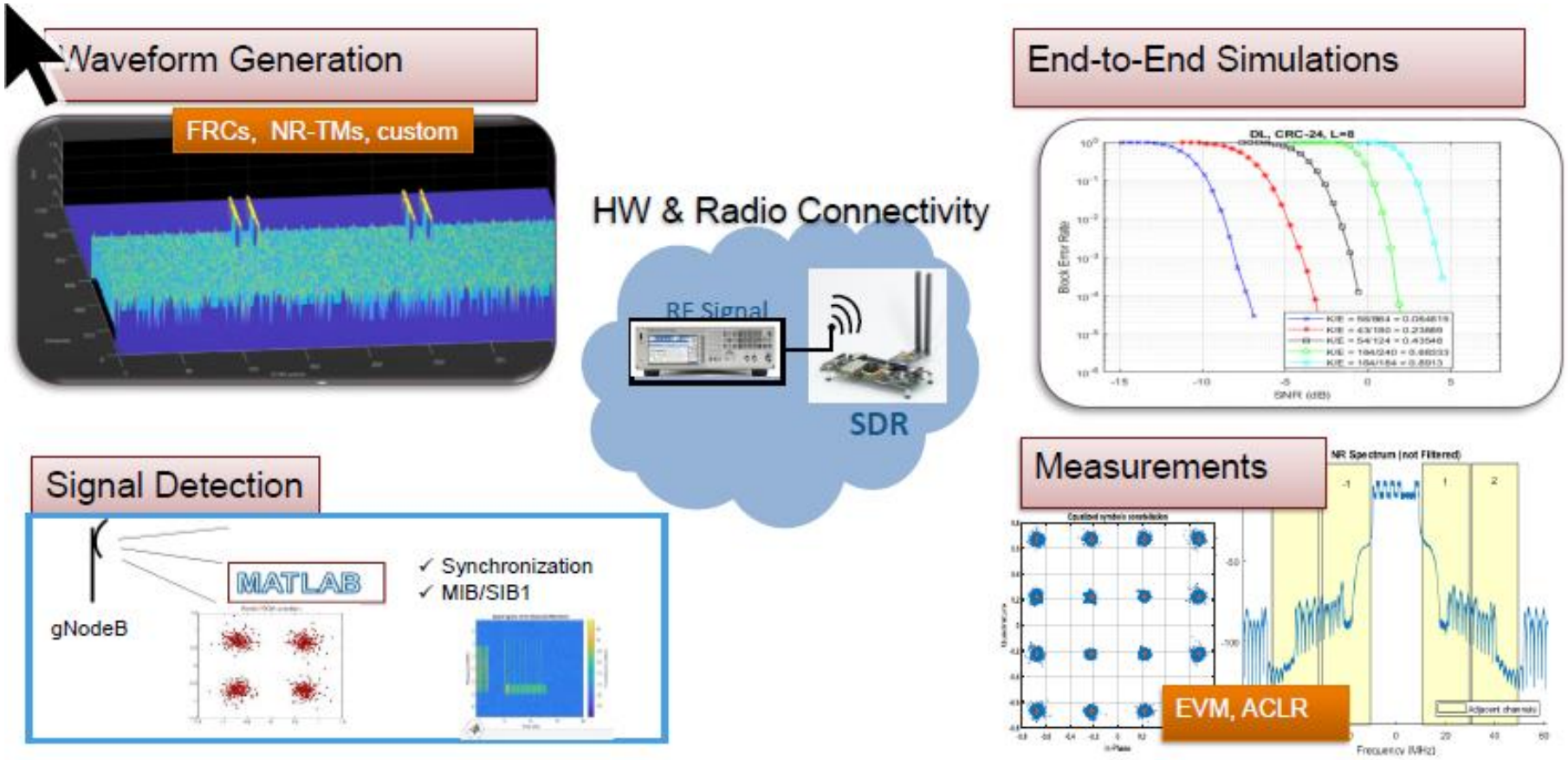




Part 3E — 5G Baseband Signal Simulation

3GPP (and More) Standard Compliant Waveform Processing

Multiple Use Cases, Beyond RF System Integration



Wireless Waveform Generation: Where to Start?

Custom and standard waveforms

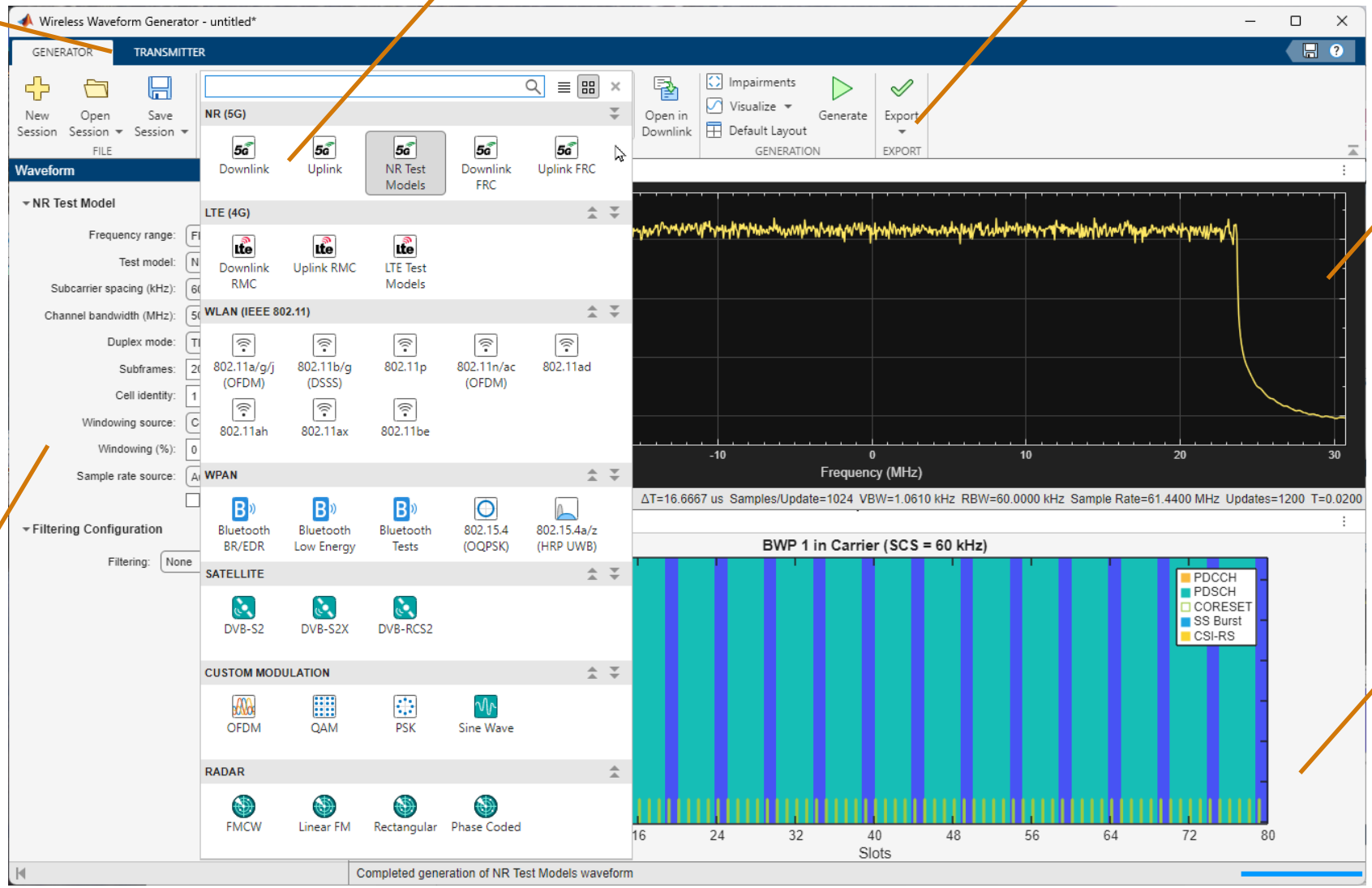
Connect to HW

Export

Spectrum

Grid allocation

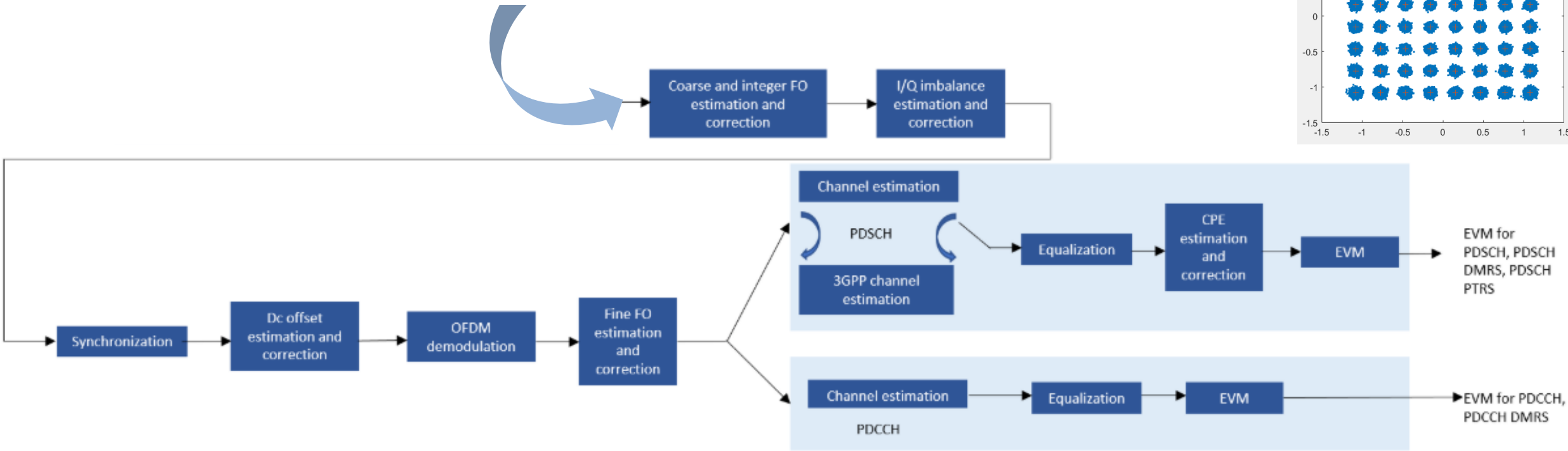
Parameters

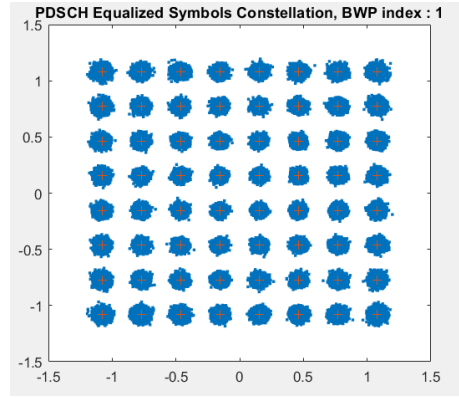


EVM Measurement Requires a Reference Receiver

Provide consistent with T&M equipment

3GPP waveform with RF impairments





Testing the RF Transmitter with 3GPP Standard Signals

Reference BB transmitter

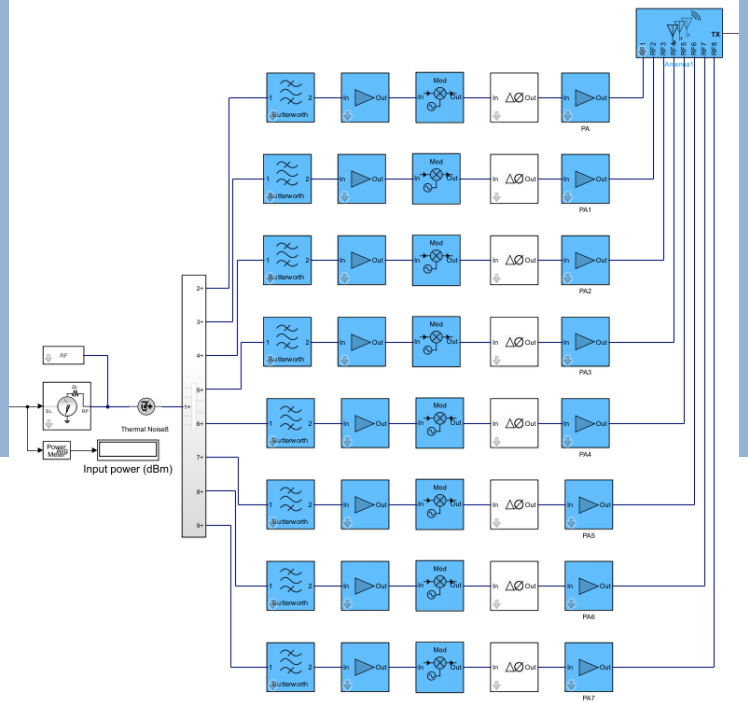
```
%% Setup for 5G 3GPP waveform
rc = "NR-FR2-TM1.1"; % Reference channel
bw = "400MHz"; % Channel bandwidth
scs = "120kHz"; % Subcarrier spacing
dm = "TDD"; % Duplexing mode
fprintf('Reference Channel = %s\n', rc);

tmwavegen = hNRReferenceWaveformGenerator(...
    rc,bw,scs,dm);
tmwavegen = makeConfigWritable(tmwavegen);
tmwavegen.Config.SampleRate = 1/Tstep;
tmwavegen.Config.NumSubframes = 1;

inWaveform = generateWaveform(tmwavegen,...
    tmwavegen.Config.NumSubframes);
```

inWaveform → **rfsystem** → outWaveform

RF Transmitter model

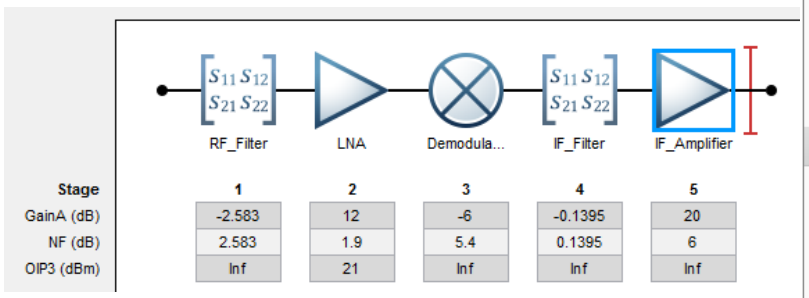


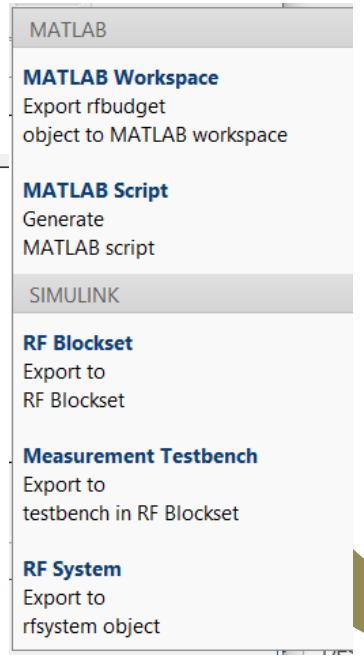
Reference BB receiver EVM/ACLR measurement

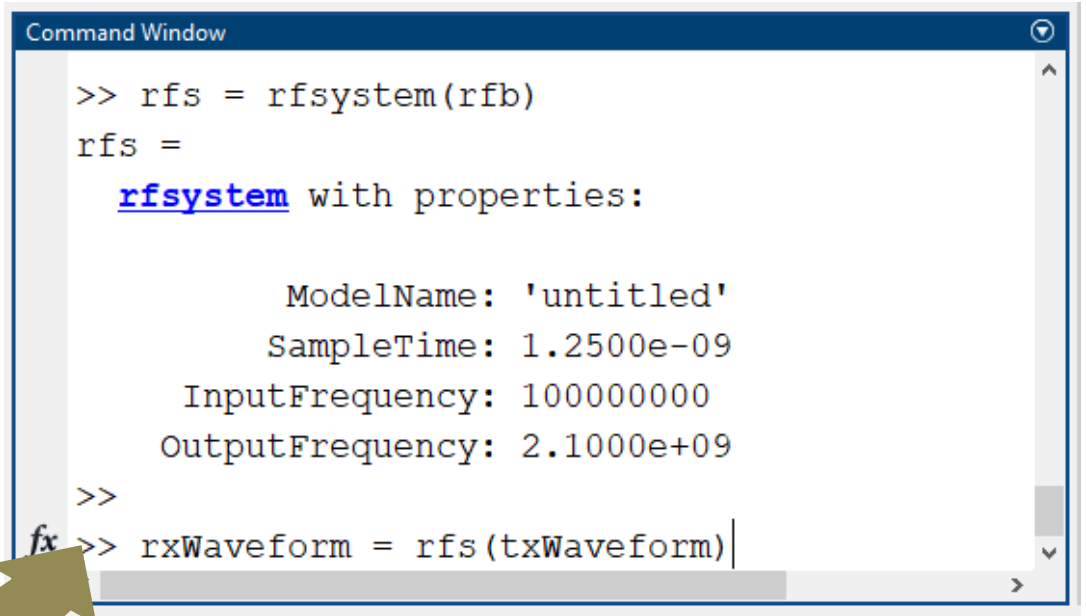
```
%% EVM Measurements
cfg = struct();
cfg.Evm3GPP = false;
cfg.TargetRNTIs = [];
cfg.PlotEVM = true;
cfg.DisplayEVM = false;
cfg.Label = tmwavegen.ConfiguredModel{1};
cfg.UseWholeGrid = true;
cfg.TimeSyncEnable = true;
cfg.CorrectCoarseFO = true;
cfg.CorrectFineFO = true;
cfg.PdcchEnable = false;
% Compute and display EVM measurements
[evmInfo,eqSym,refSym] = hNRDownlinkEVM(...
    tmwavegen.Config,outWaveform,cfg);
```

Introducing `rfsystem` Objects for Simulation in MATLAB

- Create **`rfystem`** object from RF budget
- Set up carrier frequency and simulation time step
- RF simulation in MATLAB:
  - Input signals passed as input arguments
  - Output arguments return the output signals



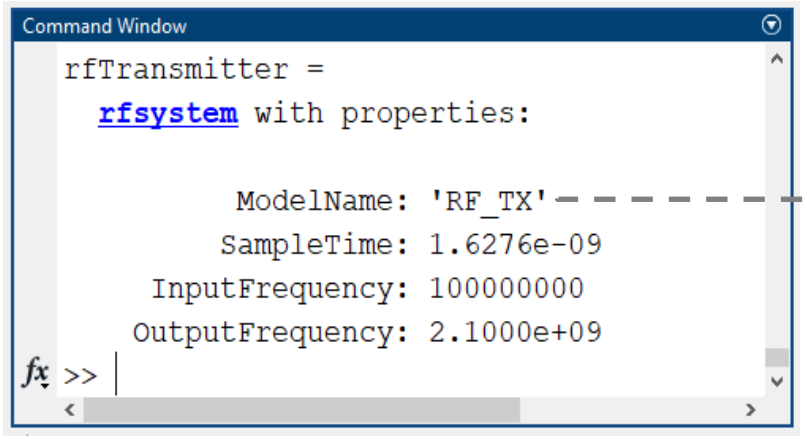


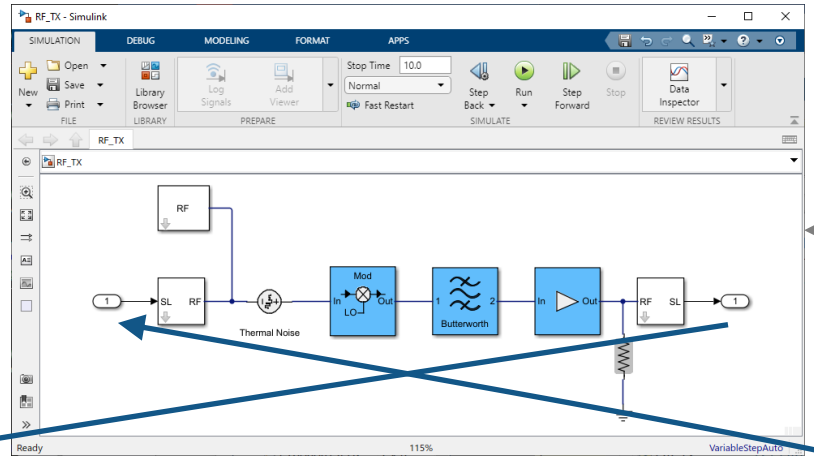


The Technology Behind `rfsystem` Objects: Circuit Envelope

Simulink model used for circuit envelope simulation

- Impedance mismatches captured in the model
- Internal states are preserved across simulations
- Perform multicarrier simulation
- Additional Simulink inputs for control signals
- Model MIMO systems






```
outSignal = rfTransmitter(inSignal)
```

Designing RF MIMO Systems is Complex!
Model Based Design Makes it Easier

- Estimate what matters: antenna coupling, dispersion, phase noise
- Scale up system design and analysis beyond textbook calculations
- Test the design using standard compliant signals and measure different metrics beyond CW
- Validate the results at every step: model component failures and spec deviation
- Gain insights into system architectures, algorithms, and scenarios before and during lab testing

